



Beggars or choosers? Lead venture capitalists and the impact of reputation on syndicate partner selection in international settings

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ABSTRACT

We investigate the importance of reputation for lead venture capitalists' choice of international and domestic syndicate partners for domestic deals. Our study is based on a novel dataset of global syndicated investments covering the period 2005 to 2014 and comprising 46,101 triads among lead investors, syndicate partners, and portfolio firms. We find reputation to be more relevant in selecting cross-border co-investors than local co-investors. The results are consistent for our international and domestic reputation scores as well as for single reputation components such as equity invested, number of investments, or number of IPO exits. In addition, we show that the reputation of the lead investor matters in international syndicate partner selection. Highly reputed venture capitalists seem to follow an intricate decision model in which they carefully choose cross-border partners over domestic ones.

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1. Introduction

Syndication in the venture capital industry helps diversify portfolios and increases the resources available to nurture portfolio firms (Brander et al., 2002; Lerner, 1994; Wilson, 1968). Syndication, however, leads to additional transaction costs and can create problems due to potentially divergent interests among syndicate partners (Cumming, 2006; Meuleman et al., 2009). In practice, partners' rights and duties cannot be fully stipulated in contracts, and even if they could, they would be difficult to enforce (Hart and Moore, 1988; Wright and Lockett, 2003; Zhelyazkov and Gulati, 2015). Therefore, venture capitalists resort to other means to reduce the risk of intra-syndicate conflicts. These include incorporating reputation as a signal for quality and reliability, along with trust, into the selection decision (Gu and Lu, 2014; Meuleman et al., 2009; Plagmann and Lutz, 2019). Furthermore, venture capitalists prefer syndicate partners similar to themselves, as like-minded partners communicate more easily and are less prone to conflict (Gompers et al., 2016; Lerner, 1994; McPherson et al., 2001).

Over the last 20 years, venture capital has become increasingly international, which exacerbates the issues involved in managing and coordinating syndicates. The number of cross-border deals, which often imply multinational syndication, has grown considerably (Aizenman and Kendall, 2012; Espenlaub et al., 2015; Tykvová and Schertler, 2014). In multinational deals, syndicate partners are not only more physically distant from each other and their investment targets but also more likely to differ in culture, language, and institutional and legal frameworks. Accordingly, information and transaction costs, as well as the risks due to augmented unfamiliarity, increase (Dai and Nahata, 2015; Tykvová and Schertler, 2014).

Drawing on the insights obtained from the reduced costs of co-operation with similar partners, we assume that foreign syndicate partners induce dissimilarity in the syndicate and thereby potentially increase the investment risks. We analyze the impact of reputation on the lead venture capitalist's selection of syndicate partners in this context of international syndication. To do so, we take into consideration a distinct scenario of internationality in syndicated venture capital deals: for lead venture capitalists investing in their home nation, we investigate the relative importance of potential domestic vs. foreign partners' reputation to the lead investor's selection decision.

We use a novel dataset consisting of approximately 46,000 triads among lead venture capitalists, partner venture capitalists, and

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their portfolio companies, covering the period from 2005 to 2014. The data stem mainly from Thomson Financial. Where necessary, we complement the dataset with country-specific data from other sources.

Previous research finds reputation to be an important signal of quality in the venture capital market (Atanasov et al., 2012; Kreps and Wilson, 1982; Shapiro, 1983). More reputable venture capitalists find it easier to raise funds and select better deals, are more successful with their investments, and are less likely to be sued for misconduct (Atanasov et al., 2012; Gompers and Lerner, 1999b; Hsu, 2004; Krishnan et al., 2011). In cases of conflict, more reputable venture capitalists have more to lose in terms of reputation (Zhelyazkov and Gulati, 2015).

Within the context of syndication, a positive reputation increases a venture capitalist's attractiveness and reduces the risks involved in cooperation, therefore enhancing desirability as a syndicate partner (Gu and Lu, 2014; Meuleman et al., 2009). Thus far, research in this area has focused on cooperation among local and foreign venture capitalists and their influence on portfolio firm performance (Bertoni and Groh, 2014; Chemmanur et al., 2016; Mäkelä and Maula, 2006, 2008). Furthermore, syndication with local syndicate partners based on similarity between lead venture capitalists and their portfolio firms has been analyzed (Dai and Nahata, 2015; Tykvová and Schertler, 2014). Meuleman et al. (2017) examine substitutes for relational embeddedness in cross-border syndication. Gu and Lu (2014) investigate the impact of reputation on syndication based on a sample of venture capital deals in China. These authors find that the impact of reputation on the likelihood of syndication differs for foreign and local (Chinese) venture capitalists. We provide further insight into these issues by analyzing the relative importance of reputation to the lead venture capitalist's syndicate partner choice. We find that the reputations of cross-border venture capitalists matter more than the reputations of local venture capitalists for the probability of cooperation in a syndicate.

Our study further adds to the literature on similarity and trust in the venture capital industry, which are interconnected. Sorenson and Stuart (2001) find that venture capitalists prefer to cooperate with syndicate partners they know from past co-investments or ones they can trust due to indirect network ties. Furthermore, Gompers et al. (2016) conclude that similarity and trust among the principals increase the likelihood of cooperation, which is also true for similarity in terms of geographic location as well as industry specialization (Trapido, 2007). We add to this literature by investigating the role of lead investor reputation in selecting syndicate partners. According to our results, however, the decision models applied by highly reputable lead venture capitalists differ from the ones used by less reputable ones.

This paper proceeds as follows. After a brief introduction, we present our theoretical foundation and develop our hypotheses. We then describe our data and methodology, followed by a presentation and discussion of our results. Finally, we set forth our conclusions.

2. Theoretical background and hypothesis development

2.1. Reputation and its importance for syndication

Syndication is a common feature of the venture capital industry. Lead venture capitalists often choose to invest jointly with partner venture capitalists in order to diversify their portfolios and increase not only the amount of the overall investment but also the resources available for selecting, monitoring, supporting, and ultimately exiting portfolio companies (Bertoni and Groh, 2014; Brander et al., 2002; Cumming et al., 2016; Lerner, 1994; Wilson, 1968).

Lead venture capitalists have a specific role within the syndicate, since it is usually they who identify the deal, choose and coordinate the syndicate partners, and monitor the investment target most closely. As the lead venture capitalist's monitoring role constitutes an important channel of information and an opportunity to intervene in the investments' management, lead venture capitalists with more frequent contact act as agents for the others and have informational advantages (Meuleman et al., 2009; Wright and Lockett, 2003). In line with the lead venture capitalists' efforts, they usually have higher equity stakes than the other equity partners (Wright and Lockett, 2003).

Syndication, as opposed to a self-contained investment, also entails transaction costs, the need for coordination, and problems due to diverging interests of the involved parties (Wright and Lockett, 2003; Zhelyazkov and Gulati, 2015). With syndication, decision-making becomes more laborious, the reduced ownership inherent in syndicated deals can abet free-riding, and conflicts among the co-investors are increasingly likely (Cumming, 2006; Meuleman et al., 2009). While syndication is widespread, as it increases the probability of success (Brander et al., 2002; Lerner, 1994; Wilson, 1968), it also leads to relatively smaller ownership stakes for each investor, which potentially reduces each partner's incentives to fulfill his or her responsibilities relative to an identical, non-syndicated investment (Wright and Lockett, 2003). Non-cooperation, or the premature withdrawal of syndicate partners, can be detrimental to the investment, and even the risk of losing potential benefits from reciprocity after the initial joint investment cannot fully prevent this (Zhelyazkov and Gulati, 2015). Syndicate partners might choose to end their participation due to loss of confidence in the investments' prospects, a capital shortage, or strategic concentration on other, more promising investments in their portfolio. Therefore, lead venture capitalists exercise great care in their selection of syndicate partners (Sorenson and Stuart, 2008; Zhelyazkov and Gulati, 2015).

Theoretically, the risks of syndication can be reduced by stipulating the partners' rights and duties in contracts or partnership agreements. In practice, however, the key aspects required for successful partnerships cannot be obligated, which necessarily leaves contracts incomplete. Since they are also difficult to enforce in practice, they are often merely used as a "backdrop" to the syndicate partnership (Wright and Lockett, 2003; Zhelyazkov and Gulati, 2015).

Reputation is often used as an informal enforcement mechanism in such settings. Due to obstacles in contract enforceability, defecting partners usually fear reputational consequences more than legal ones (Zhelyazkov and Gulati, 2015). Defection and early withdrawal from syndicates disrupt relationships and undermine a partner's reputation. In the tightly networked venture capital industry, this reduces chances of renewed syndication and further deal flow participation (Wright and Lockett, 2003; Zhelyazkov and Gulati, 2015). As Atanasov et al. (2012) have shown, lawsuits against venture capitalists significantly reduce their future business. More reputable venture capitalists have been shown to be less likely to be sued for misconduct, as they have more to lose and behave accordingly (Atanasov et al., 2012; Meuleman et al., 2009; Petkova et al., 2014).

Furthermore, reputation signals a potential syndicate partner's experience, network status within the industry, and track record in successful deal selection, management, and exit (Gu and Lu, 2014; Krishnan et al., 2011; Meuleman et al., 2009). A good reputation increases a potential partner's attractiveness for syndication, as the syndicate hopes to gain access to the reputable partner's valuable resources and benefit from its expertise by partnering up (Gu and Lu, 2014; Meuleman et al., 2009).

Table 1.1

Distribution of lead venture capitalists' domestic deals and their partners' nationalities (realized cases).

LEAD nation	PARTNER nation																							
	AUS	BEL	CAN	CHN	DNK	FIN	FRA	DEU	IND	IRL	ISR	ITA	JPN	KOR	NLD	NOR	RUS	SGP	ESP	SWE	CHE	GBR	USA	Sum
BEL		1																						1
CAN			43							1	1										2	3	19	69
CHN				105							2		1	1				5			1	1	24	140
FRA							92															1	93	
DEU								24								2	1						2	31
IND									4											1	1		3	7
ISR											3							1					2	6
JPN													1											1
KOR														2										2
NLD															5									5
ESP																			3					3
SWE																				15				15
GBR							3					1	1		1				3		2	56	12	79
USA	5		29	4	0	0	1	10	4	5	13		3	3	5	5	3		2	3	27	24	3593	3739
Sum	5	1	72	109	0	0	96	34	8	6	19	1	6	6	11	7	4	6	8	18	33	86	3655	4191

Note: This table represents the distribution of realized deals by lead investors in their home countries and the nationalities of their syndicate partners in our dataset. The three-letter country codes are ISO-alpha-3 codes and refer to the following nations: AUS=Australia; BEL=Belgium; CAN=Canada; CHN=China; DNK=Denmark; FIN=Finland; FRA=France; DEU=Germany; IND=India; IRL=Ireland; ISR=Israel; ITA=Italy; JPN=Japan; KOR=Korea; NLD=Netherlands; NOR=Norway; RUS=Russia; SGP=Singapore; ESP=Spain; SWE=Sweden; CHE=Switzerland; GBR=United Kingdom; USA=United States of America.

Table 1.2

Distribution of lead venture capitalists' domestic deals and their partners' nationalities (matched cases).

LEAD nation	PARTNER nation																							
	AUS	BEL	CAN	CHN	DNK	FIN	FRA	DEU	IND	IRL	ISR	ITA	JPN	KOR	NLD	NOR	RUS	SGP	ESP	SWE	CHE	GBR	USA	Sum
BEL		10																						10
CAN		3	430				4	5						7	6						4	27	204	690
CHN			1	1050			9	3			8		12	17	2		4	93		8		19	174	1400
FRA		2					920			1								1				4	2	930
DEU	1	5	1	1		1	3	240							9	2	4			3	17	11	12	310
IND			2					3	40						2			7	2				14	70
ISR								3			30	3	2									3	19	60
JPN													10											10
KOR														20										20
NLD															50									50
ESP																			30					30
SWE																				150				150
GBR	2	11	7	2	7	5	8	14		6	2	3	2		15	11	4		5	4	5	560	117	790
USA	97	28	202	43	20	11	66	135	26	40	163	6	54	12	38	36	45	44	13	35	96	250		35,930
Sum	100	59	643	1096	27	17	1010	403	66	46	204	12	80	56	122	49	57	145	50	200	122	874		36,472
																								41,910

Note: This table represents the distribution of realized deals by lead investors in their home countries and the nationalities of the matched syndicate partners in our dataset. As we matched 10 partners to each realized case, our full dataset comprising realized cases and controls contains the numbers given here and in Table 1.1. The three-letter country codes are ISO-alpha-3 codes and refer to the following nations: AUS=Australia; BEL=Belgium; CAN=Canada; CHN=China; DNK=Denmark; FIN=Finland; FRA=France; DEU=Germany; IND=India; IRL=Ireland; ISR=Israel; ITA=Italy; JPN=Japan; KOR=Korea; NLD=Netherlands; NOR=Norway; RUS=Russia; SGP=Singapore; ESP=Spain; SWE=Sweden; CHE=Switzerland; GBR=United Kingdom; USA=United States of America.

2.2. Syndication in an international context

While the specific characteristics of venture capital make it best suited for short-distance investments, the industry has become increasingly international in recent decades (Chen et al., 2010; Cumming and Dai, 2010; Kräussl et al., 2015). Monitoring, support, and oversight, which are required for adding value in portfolio companies, are more costly when venture capitalists and their portfolio companies are located far from each other (Chen et al., 2010). Still, increased competition for promising investment targets due to more demand for the financing instrument, as well as regulatory improvements, has led venture capitalists to invest outside their home markets (Devigne et al., 2013).

Where venture capitalists invest abroad, they suffer from the liability of foreignness, meaning they have comparatively less knowledge about local markets, habits, and customs and are less connected via local networks (Gu and Lu, 2014; Meuleman et al., 2017; Tykvová and Schertler, 2014). Therefore, foreign venture capi-

talists often syndicate with a local partner (Dai and Nahata, 2015). Syndicates consisting of both foreign and local venture capitalists have a higher probability of successful exits, both via trade sales and via initial public offerings (Bertoni and Groh, 2014; Cumming et al., 2016; Humphery-Jenner and Suchard, 2013).

Furthermore, local venture capitalists may opt for foreign syndicate partners strategically when the investment requires specific expertise or resources that are not available among potential local co-investors (Chemmanur et al., 2016; Devigne et al., 2016; Devigne et al., 2013; Tykvová and Schertler, 2011). They might also invite syndicate partners from specific countries in order to facilitate the internationalization of their portfolio companies in these markets or to prepare an international exit (Humphery-Jenner and Suchard, 2013; LiPuma and Park, 2014).

Syndication among partners from different nationalities, however, also increases the syndicate's heterogeneity. This dissimilarity stems from national differences in terms of institutional and legal setups and the development and size of financial and venture capi-

Table 2
Definition of variables.

Variables	Definition
Dependent variable: COOPERATION	A binary variable that equals one for a triad that contains the partner selected to a venture capital syndicate and zero otherwise.
Independent variables:	
PARTNER_int_rep	A metric variable that depicts a reputation score calculated from five components, namely, the partner VC's age, the average number of funds managed over the prior ten years, the equity invested over the last ten years, the number of start-ups invested outside the respective venture capitalist's domestic market over the last ten years, and the number of start-ups taken public over that period. We follow the methodology of Lee et al. (2011) in calculating the index.
PARTNER_dom_rep	A metric variable that depicts a reputation score calculated from five components, namely, the partner VC's age, the average number of funds managed over the prior ten years, the equity invested over the last ten years, the number of start-ups invested in within the respective venture capitalist's domestic market over the last ten years, and the number of start-ups taken public over that period. We follow the methodology of Lee et al. (2011) in calculating the index.
PARTNER_equityinv	A metric variable representing the total dollar amount of equity a partner VC has invested over the course of five years prior to the actual investment date.
PARTNER_int_pfc	A metric variable counting the absolute number of portfolio companies a partner VC has invested in outside its domestic market in the ten years prior to the actual investment date.
PARTNER_dom_pfc	A metric variable counting the absolute number of portfolio companies a partner VC has invested in within its domestic market in the ten years prior to the actual investment date.
PARTNER_fundsmgd	A metric variable reporting the average number of funds managed over the course of ten years prior to the actual investment date.
PARTNER_age	A metric variable counting the number of months between the first reported investment since 1990 and the actual investment date.
PARTNER_fundsmgd	A metric variable reporting the average number of funds managed over the course of ten years prior to the actual investment date.
PARTNER_ipos	A metric variable reporting the total number of reported IPOs a partner VC achieved over the course of the ten years prior to the actual investment date.
Control variables:	
LEAD_PARTNER_cultdist4	A metric variable summarizing the Cartesian distance of the four main cultural dimensions developed by Hofstede (1980), power distance, individualism, masculinity, and uncertainty avoidance, among lead and partner VCs, based on the following formula: $LEAD_PARTNER_cultdist4 = \frac{\sum_{i=1}^4 (CulturalDimension_{LEAD,i} - CulturalDimension_{PARTNER,i})^2}{4} \cdot 0.5$
LEAD_PARTNER_geodist	A metric variable that equals the geodetic distance between the headquarters of the lead and partner VCs in miles.
PARTNER_target_indexp5	An ordinal variable for the partner's experience in the target's (i.e., portfolio company's) industry. It relates the partner's major industry focus (i.e., the industry into which the partner has made most of its investments over the previous five years) to the industry in which the portfolio company operates. The values range from 0 (the partner's major industry is a different primary industry group) to 3 (the partner's major industry is the same minor industry group). We use the Venture Economic Industry Codes (VEIC), which have been developed by Thomson Financial.
LEAD_int_reputation	A metric variable that depicts a reputation score calculated from five components, namely, the lead VC's age, the average number of funds managed over the prior ten years, the equity invested over the last ten years, the number of start-ups invested in outside the respective VC's domestic market over the last ten years, and the number of start-ups taken public over that period. We follow the methodology of Lee et al. (2011) in calculating the index.
Round_synsize	A metric variable that represents the number of partners in the original syndicate the realized lead-partner dyad stems from.
Target_Market_Change_t3	A metric variable that relates the number of deals in the target market in the current year to the number of deals three years earlier ($t/(t-3)$).
Target_stage	An ordinal variable that depicts the portfolio company's stage. It ranges from 1 to 4, with the following values: 1 (seed), 2 (early stage), 3 (later stage), 4 (expansion).
Target_industry_FE	A set of binary variables for each primary industry category in our dataset. They equal one if the portfolio company operates in that specific category and zero otherwise. We use the Venture Economic Industry Codes (VEIC)®, which have been developed by Thomson Financial.
LEAD_nation_FE	A set of binary variables for each of 23 lead nations in our sample. They equal one for a lead from that nation and zero otherwise.
Inv_year_FE	A set of binary variables for each year within our period of observation (2005–2014). They equal one for an investment in that specific year and zero otherwise.

Note: This table displays the measurement of all variables used in our econometric models. To analyze the characteristics that have an impact on the probability of cooperation among lead and potential partner VCs, we use a binomial logit model.

tal markets, as well as differences in culture (Dai and Nahata, 2015; La Porta et al., 1998; Nahata et al., 2014). Each nation's specific institutions shape the goals and actions of its people, which can create substantial heterogeneity in group and individual behavior (Bruton and Ahlstrom, 2003; North, 1990). For example, it has been documented that firms from Asian economies are bound more by social networks and reciprocal exchanges than those from western economies, for which adherence to regulations and contracts is much more important (Li et al., 2008). Differences in value systems can produce diverging expectations and precipitate a lack of commitment and conflict (Lane and Beamish, 1990). Due to the heterogeneity introduced by multinational syndicate members, the typical problems of asymmetric information and agency issues within syndicates are aggravated.

Against this backdrop of advantages and disadvantages of cross-border syndication, our aim is to understand the relevance of syndicate partner reputation to the probability of cooperation in a distinct international setting. We focus on lead investors located in the same nation as the investment target and that select partners to join the syndicate from abroad or locally.

2.3. Local lead venture capitalists and the impact of reputation on their choice of international syndicate partners

Where the lead venture capitalist invests domestically and is located in the country of the portfolio company, he or she has a choice among local partners from the same nation or partners from abroad. Cross-border relationships entail particular risks for cooperation as they induce dissimilarity in the syndicate partnership and thereby render its management and the transfer of information among the partners more complex than for homogeneous syndicates consisting of domestic partners only (Gompers et al., 2016; McPherson et al., 2001). Furthermore, the physical distance between foreign syndicate partners and the portfolio company makes it more difficult for them to become effectively involved in the investment target's oversight, support, and management, since foreign partners are more difficult to approach (Cumming, 2006; Dai et al., 2012). Finally, the higher risk involved in cross-border investments potentially leads to lengthier discussions and more complex decision-making processes among the syndicate partners (Meuleman et al., 2009). Lead venture capital-

Table 3
Descriptive statistics.

Variables	Obs.	Mean	S.D.	Min.	Max.
Dependent variable: COOPERATION	46,101	0.0909	0.287	0	1
Independent variables:					
PARTNER_int_rep	46,101	7.173	8.977	0.0166	100
PARTNER_dom_rep	46,101	8.444	10.25	0.0162	100
PARTNER_equityinv	46,101	548.7	1231	0	15,920
PARTNER_int_pfc	46,101	2.714	10.62	0	421
PARTNER_dom_pfc	46,101	23.18	32.78	0	369
PARTNER_age	46,101	10.40	9.448	0	57
PARTNER_fundsmgd	46,101	1.342	1.434	0	12.30
PARTNER_ipos	46,101	2.381	5.912	0	90
PARTNER_local	46,101	0.942	0.234	0	1
Control variables:					
LEAD_PARTNER_cultdist4	46,101	0.598	2.877	0	25.26
LEAD_PARTNER_geodist	46,101	1743	1687	0	10,635
PARTNER_target_indexp5	46,101	1.102	1.208	0	3
LEAD_int_reputation	46,101	13.65	14.91	0.0166	100
Round_synsize	46,101	3.165	1.399	2	11
Target_Market_Change_t3	46,101	1.248	0.599	0.0628	18.15
Target_stage	46,101	2.449	1.010	1	4

Note: This table displays summary statistics for all variables used in our models (except the dummy variables to control for investment year and target industry).

Table 4
Variance inflation factors and correlation coefficients.

Variables	VIF	1	2	3	4	5	6	7	8	9	10	11
1 COOPERATION		1.000										
2 PARTNER_int_rep		0.123*	1.000									
3 PARTNER_dom_rep	1.03	0.131*	0.967*	1.000								
4 PARTNER_local	3.67	0.000	−0.028*	0.036*	1.000							
5 LEAD_PARTNER_cultdist4	3.69	−0.002	0.026*	−0.028*	−0.836*	1.000						
6 LEAD_PARTNER_geodist	1.30	−0.062*	0.034*	0.009*	−0.389*	0.391*	1.000					
7 PARTNER_target_indexp5	1.12	0.211*	0.016*	0.039*	0.038*	−0.030*	−0.032*	1.000				
8 Target_Market_Change_t3	5.17	0.000	−0.021*	−0.062*	−0.167*	0.203*	0.041*	−0.023*	1.000			
9 Target_stage	1.10	0.000	−0.017*	−0.035*	−0.047*	0.044*	−0.026*	−0.010*	0.078*	1.000		
10 LEAD_int_reputation	1.05	0.000	0.032*	0.028*	0.023*	−0.023*	0.064*	−0.007	−0.058*	−0.075*	1.000	
11 Round_synsize	1.08	0.000	−0.011*	0.006	0.016*	−0.019*	0.014*	0.022*	−0.029*	−0.079*	0.044*	1.000
Target_industry_FE		Incl.	Incl.	Incl.	Incl.	Incl.	Incl.	Incl.	Incl.	Incl.	Incl.	Incl.
LEAD_nation_FE		Incl.	Incl.	Incl.	Incl.	Incl.	Incl.	Incl.	Incl.	Incl.	Incl.	Incl.
Inv_year_FE		Incl.	Incl.	Incl.	Incl.	Incl.	Incl.	Incl.	Incl.	Incl.	Incl.	Incl.
Mean VIF	2.13											

Note: This table shows variance inflation factors (VIFs) and pairwise correlations for the variables of key interest. Significance is marked by * at 5%.

ists are therefore likely to choose homogeneous, domestic partners (Gompers et al., 2016; McPherson et al., 2001; Wright and Lockett, 2003).

Lead venture capitalists, however, might need specific resources for their investments. These could entail specifically required industry expertise; an objective, unbiased opinion on when to write off an unviable investment; or access to foreign exit markets and knowledge on how to prepare an international exit (Bertoni and Groh, 2014; Cumming et al., 2016; Humphery-Jenner and Suchard, 2013; LiPuma and Park, 2014). Where these resources are scarce or unavailable locally, lead venture capitalists might specifically look for partners abroad (Chemmanur et al., 2016; Devigne et al., 2016; Devigne et al., 2013; Hochberg et al., 2015; Tykvová and Schertler, 2011).

Recognizing the additional risk of inducing dissimilarity in the syndicate partnership, lead venture capitalists are likely to place higher emphasis on foreign partners' reputations in order to safeguard themselves against the risk introduced by partners whose actions are more difficult to predict than those of local partners (Wright and Lockett, 2003). We therefore derive the following hypothesis:

Hypothesis 1. In their selection of syndicate partners, lead venture capitalists place more emphasis on the reputations of foreign partners than on the reputations of domestic ones.

A venture capitalist's local and international reputations likely differ. Venture capital has conventionally been a short-distance business characterized by tight networks among local stakeholders, in which news travels fast and information with reputational value is quickly shared. Internationally, these networks are looser, which renders international reputation harder to establish (Gu and Lu, 2014; Meuleman et al., 2017; Tykvová and Schertler, 2014). Most likely, international reputation differs from the reputation in the local, domestic market. The success of potential co-investors on the international rather than domestic level is more likely to be recognized by lead investors located outside of the home market of the syndicate partner candidate. International investment activity adds credibility to an investor's ability to successfully operate outside its home market and add value to investments abroad. The relative standing of a venture capitalist in its home market, however, might also be an important component of its attractiveness as a syndicate partner. Especially in cases where foreign partners are sought

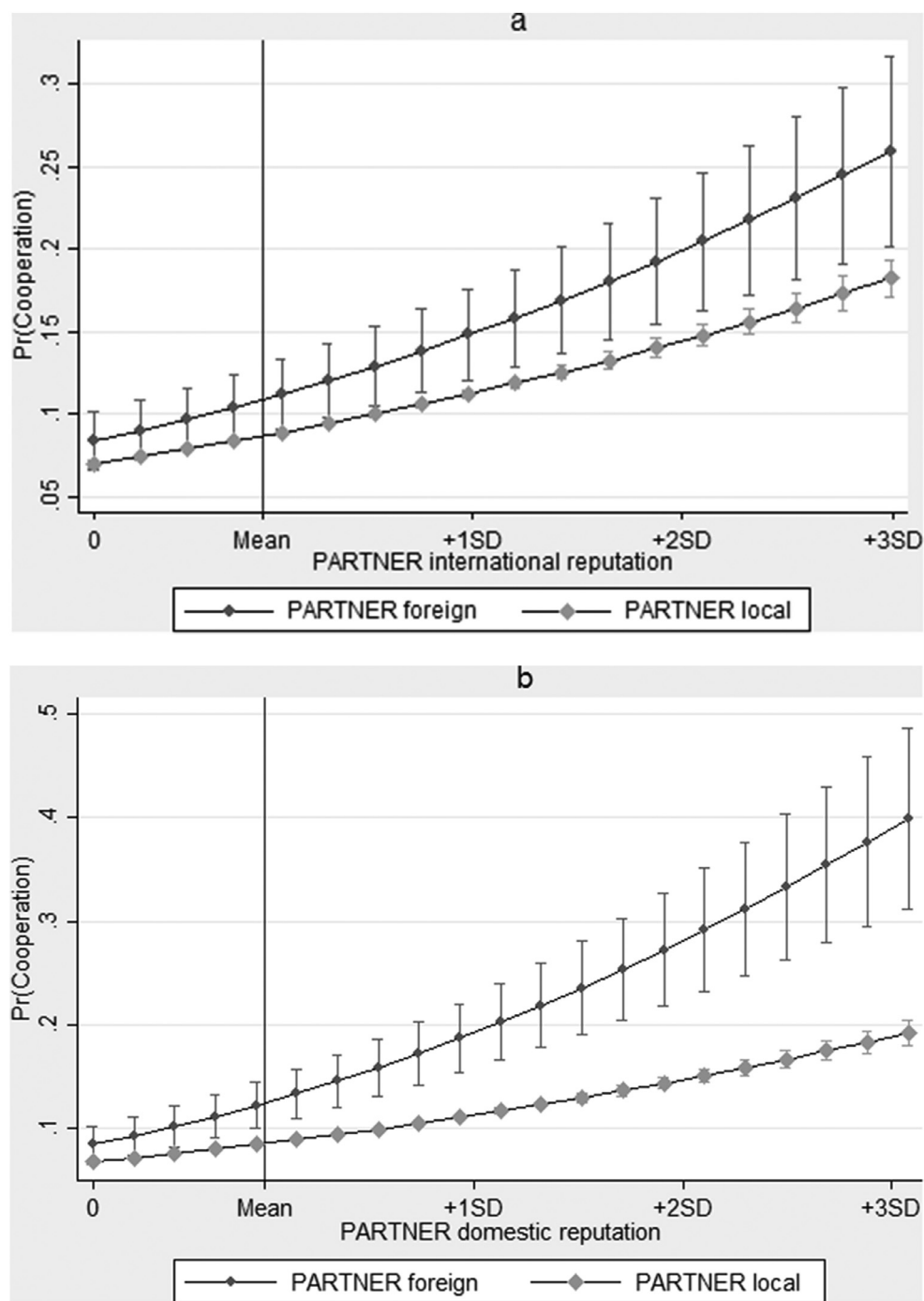


Fig. 1. Impact of potential local vs. foreign partners' international and domestic reputations on the probability of cooperation.

Note: Figs. 1(a) and 1(b) present the calculated predictions of Model 1 and Model 2 (Table 5). We show predictive margins with 95% confidence intervals for foreign and local partners, respectively, depending on the values of PARTNER_international_reputation (Fig. 1(a)), and PARTNER_domestic_reputation (Fig. 1(b)).

due to their expertise in their home market (for example, in order to prepare for an IPO in that country) these partners might be attractive due to being the most reputable venture capitalist in their home market even without an international reputation. Therefore, a venture capitalist's domestic and international reputations must be differentiated in an international framework. For both types of reputation, we expect the lead venture capitalists to value the reputations of foreign partners higher than of domestic ones.

2.4. Beggars or choosers – the impact of lead venture capitalists' own reputations on their choice of syndicate partners

Whether lead venture capitalists are able to select syndicate partners based on their preferences with regard to the partners' reputations is likely influenced by their bargaining position, one important component of which is their own reputation. Previous research has already shown that private equity firms' reputations improve their bargaining position and help them negotiate more

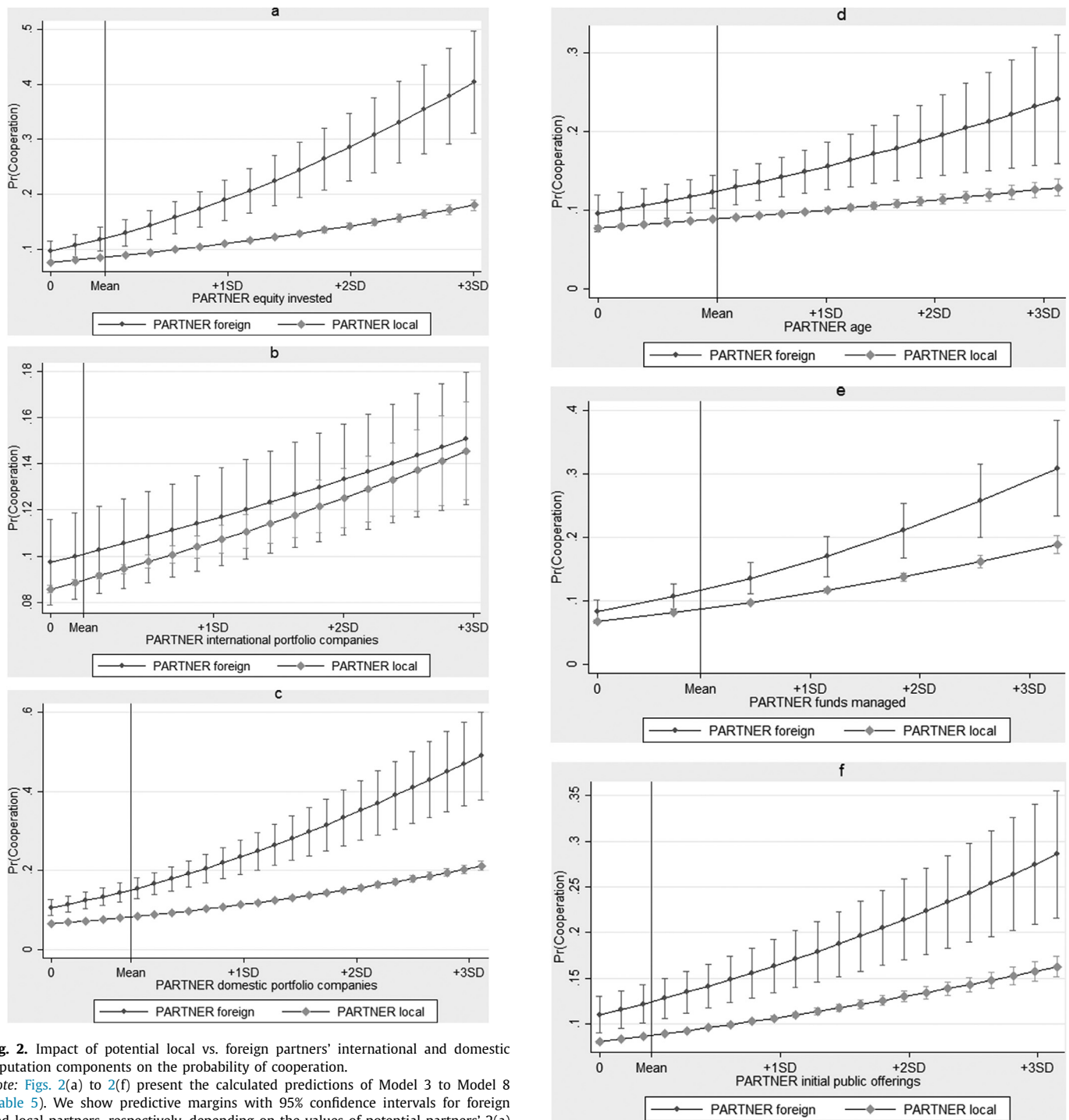


Fig. 2. Continued

Fig. 2. Impact of potential local vs. foreign partners' international and domestic reputation components on the probability of cooperation.

Note: Figs. 2(a) to 2(f) present the calculated predictions of Model 3 to Model 8 (Table 5). We show predictive margins with 95% confidence intervals for foreign and local partners, respectively, depending on the values of potential partners' 2(a) equity invested, 2(b) number of investments in international portfolio companies, 2(c) number of investments in domestic portfolio companies, 2(d) age, 2(e) number of funds managed and 2(f) total number of IPOs.

favorable contracts (Demiroglu and James, 2010), achieve a lower valuation of their portfolio companies (Hsu, 2004), and offset the agency problems of syndication (Meuleman et al., 2009).

We have already argued that lead venture capitalists likely turn to foreign syndicate partners due to a lack of required skills and resources among potential syndicate partners from their own nation (Chemmanur et al., 2016; Devigne et al., 2016; Devigne et al., 2013; Hochberg et al., 2015; Tykrová and Schertler, 2011). Since more reputable lead venture capitalists have greater bargaining power, we assume they can place more emphasis on cross-border partners' reputations when the need to turn to a foreign partner arises.

For lead venture capitalists with a low-to-medium reputation, that might not be the case, as these venture capitalists do not stand out as particularly attractive leads for co-investment. They have relatively low bargaining power to begin with, as lead venture capitalists are the ones who need to find a partner with particular skills or resources. Accordingly, we assume lead venture capitalists with a low-to-medium reputation to be beggars rather than choosers and those with high reputation to be able to choose more reputable foreign partners to offset the disadvantages for syndication management introduced by multinationality. Therefore, we formulate the following second hypothesis:

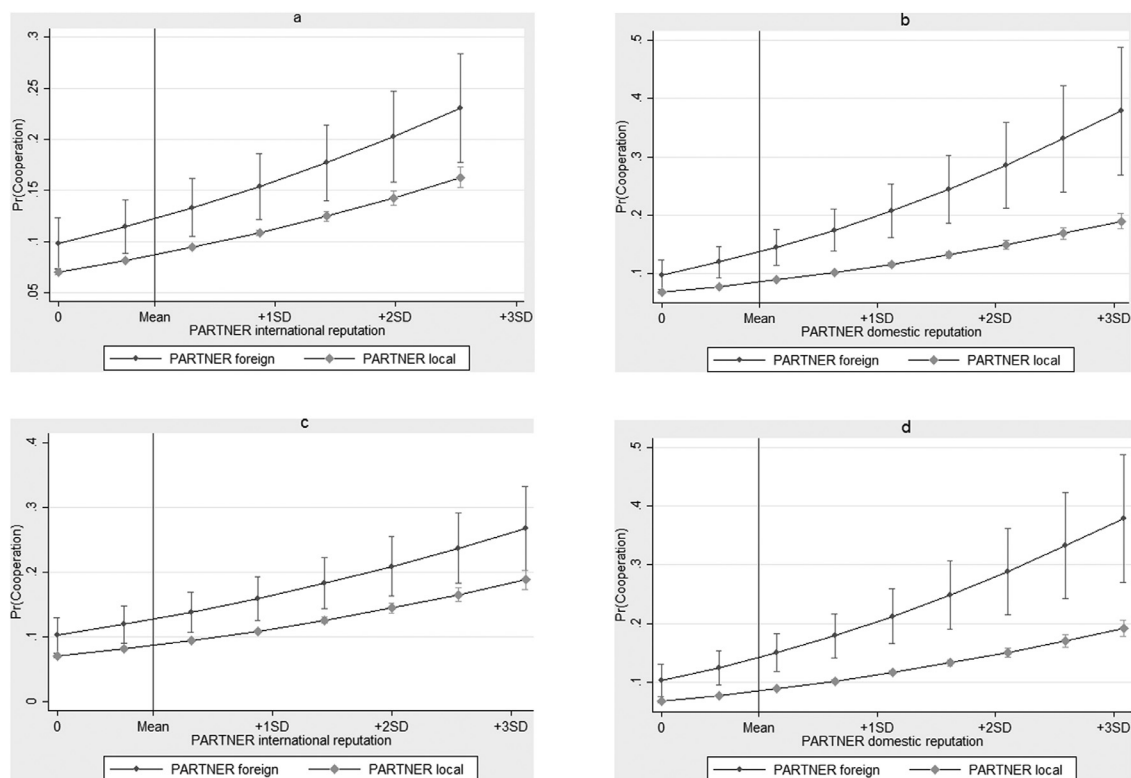


Fig. 3. Impact of potential local vs. foreign partners' reputations on the probability of cooperation for subsamples of low to medium lead investor reputation.

Note: Figs. 3(a) and 3(b) are based on a subsample of triads including only lead investors with low to medium international reputation. Figs. 3(c) and 3(d) are based on a subsample of triads including only lead investors with low to medium domestic reputation. The figures present calculated predictions of Model 1 to Model 4 (Table 6). We show predictive margins with 95% confidence intervals for foreign and local partners, respectively, depending on the values of PARTNER_international_reputation (Fig. 3(a) and Fig. 3(c)), and PARTNER_domestic_reputation (Fig. 3(b) and Fig. 3(d)).

Hypothesis 2. In their selection of syndicate partners, lead venture capitalists who themselves command a high reputation can ask for a higher reputation premium to offset the additional risks of cross-border syndicate partners than can those lead venture capitalists with a low-to-medium reputation.

3. Data and methodology

3.1. Data sources

Our research is based on data from the Private Equity Module of ThomsonONE Investment Banking (formerly VentureXpert) provided by Thomson Financial. This module contains information on global venture capital deals. Our dataset comprises data on national and multinational syndicates with venture capitalists from 14 nations for the period of 2005–2014 (see Tables 1.1–1.2 for a distribution of deals by lead and partner nationalities). We augment the resulting database with venture capital market data from Thomson Financial and data on Hofstede's cultural dimensions.¹

3.2. Empirical strategy and structure of the dataset

To avoid any bias resulting from the differing objectives of various types of venture capital providers (i.e. government, corporate venture capital, etc.), we analyze the deals of independent venture capital firms only. As we investigate the impact of reputation on syndicate partner selection, we restrict the sample to deals

in which a specific lead venture capitalist cooperates with a partner for the first time. Therefore, we can safely assume that mutual familiarity from previous joint investments does not affect the probability of cooperation or confuse reputation with further information obtained through previous interaction. By analyzing cooperation at the venture capital firm level (as opposed to joint investments of venture capital funds), we ensure that our analysis is unbiased by co-investments with affiliated funds, which would violate the independence assumption of our selection decisions (Cumming et al., 2010). We further restrict our analysis to the first venture capital investment in a specific portfolio company, or, where the first investment was a non-syndicated one, to the first syndicated investment round of a specific portfolio company. This is necessary in order to ensure that potential path dependencies of previous syndication do not affect the probability of cooperation in the round we analyze. That way, our data are not affected by potential switches of lead investors, which might occur over the course of several investment rounds (Cumming and Dai, 2013).

In our analysis of relationship formation among venture capitalists, we follow Sorenson and Stuart (2008) and focus on the probability of cooperation between the lead venture capitalist and a potential syndicate partner given a specific investment target. We model the lead venture capitalist's choice of a specific syndicate partner and therefore cannot account for the simultaneity of syndicate partner choices. Since the lead venture capitalists are the ones who search for potential syndicate partners and approach them with an offer to co-invest (Wright and Lockett, 2003), however, we cover the major channel in initiation of a syndicate even though we cannot model the partners' decisions of whether or not to accept an offer. While we analyze the relevance of the partners'

¹ Taken from Geert Hofstede's website (<http://geerthofstede.com>).

Table 5

Empirical analysis: The relative impact of reputation and various components on the probability of cooperation in a syndicate.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Dependent variable:	Cooperation	Cooperation	Cooperation	Cooperation	Cooperation	Cooperation	Cooperation	Cooperation
Independent variables and interactions:								
PARTNER_int_rep	0.0435*** (0.00519)							
PARTNER_int_rep x PARTNER_local	−0.00883 (0.00548)							
PARTNER_dom_rep		0.0550*** (0.00589)						
PARTNER_dom_rep x PARTNER_local		−0.0228*** (0.00608)						
PARTNER_equityinv			0.000487*** (5.18e-05)					
PARTNER_equityinv x PARTNER_local			−0.000237*** (5.31e-05)					
PARTNER_int_pfc				0.0159*** (0.00232)				
PARTNER_int_pfc x PARTNER_local				0.00298 (0.00381)				
PARTNER_dom_pfc					0.0189*** (0.00215)			
PARTNER_dom_pfc x PARTNER_local					−0.00718*** (0.00219)			
PARTNER_age						0.0302*** (0.00827)		
PARTNER_age x PARTNER_local						−0.0151* (0.00846)		
PARTNER_fundsmgd							0.293*** (0.0389)	
PARTNER_fundsmgd x PARTNER_local							−0.0837** (0.0407)	
PARTNER_ipos								0.0621*** (0.00903)
PARTNER_ipos x PARTNER_local								−0.0210** (0.00939)
PARTNER_local	−0.207 (0.129)	−0.259** (0.131)	−0.276** (0.122)	−0.148 (0.119)	−0.566*** (0.118)	−0.245 (0.151)	−0.236* (0.135)	−0.373*** (0.115)
Control variables:	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
LEAD_PARTNER_cultdist4	0.0223* (0.0116)	0.0177 (0.0115)	0.0196* (0.0117)	0.0262** (0.0114)	0.00603 (0.0113)	0.0213* (0.0111)	0.0217* (0.0114)	0.0153 (0.0112)
LEAD_PARTNER_geodist	−0.000172*** (1.34e-05)	−0.000172*** (1.34e-05)	−0.000172*** (1.36e-05)	−0.000175*** (1.37e-05)	−0.000169*** (1.34e-05)	−0.000174*** (1.36e-05)	−0.000175*** (1.35e-05)	−0.000169*** (1.34e-05)
PARTNER_target_indepx5	0.676*** (0.0151)	0.677*** (0.0152)	0.675*** (0.0152)	0.669*** (0.0149)	0.673*** (0.0155)	0.671*** (0.0148)	0.674*** (0.0150)	0.673*** (0.0150)
LEAD_int_reputation	0.000341 (0.000410)	0.000338 (0.000412)	0.000324 (0.000411)	0.000671* (0.000392)	0.000350 (0.000435)	0.000648* (0.000379)	0.000451 (0.000402)	0.000397 (0.000401)
Round_syssize	−0.00198 (0.00465)	−0.00153 (0.00469)	−0.00267 (0.00472)	−0.00666 (0.00452)	−0.00336 (0.00475)	−0.00581 (0.00440)	−0.00204 (0.00468)	−0.00356 (0.00457)
Target_Market_Change_t3	0.0690* (0.0359)	0.0575 (0.0373)	0.0773** (0.0380)	0.0969*** (0.0331)	0.0677* (0.0403)	0.0837** (0.0356)	0.0719** (0.0348)	0.0952** (0.0388)
Target_stage	−0.00230 (0.00605)	−0.00180 (0.00610)	−0.000615 (0.00620)	0.000373 (0.00563)	0.000560 (0.00646)	−0.00122 (0.00547)	−0.000905 (0.00585)	−0.00355 (0.00583)
Target_industry_FE	Yes	yes	yes	yes	yes	yes	yes	yes
LEAD_nation_FE	Yes	yes	yes	yes	yes	yes	yes	yes
Inv_year_FE	Yes	yes	yes	yes	yes	yes	yes	yes
Constant	−2.711*** (0.130)	−2.728*** (0.130)	−2.567*** (0.124)	−2.555*** (0.124)	−2.357*** (0.118)	−2.584*** (0.148)	−2.745*** (0.135)	−2.352*** (0.118)
Observations	46,101	46,101	46,101	46,101	46,101	46,101	46,101	46,101
Pseudo R-squared	0.106	0.109	0.111	0.0935	0.119	0.0900	0.102	0.0997

Note: In this table, we present results of binomial logistic regressions with the probability of cooperation between lead investor and co-investors in a syndicate as the dependent variable. As independent variables, we use various metric proxies for syndicate partner reputation and interactions terms thereof with the dummy variable PARTNER_local, which equals one when the syndicate partner is located in the same country as the portfolio company and zero otherwise. Heteroskedasticity robust standard errors are reported in parentheses, significance is marked by * at 10%, ** at 5%, and *** at 1%. Standard errors are clustered around portfolio firms. The interaction terms of Model 1 and Model 2 are displayed graphically in Fig. 1(a/b), and the interaction terms of Models 3 to 8 are displayed in Fig. 2(a to f).

reputations as a determinant of the lead venture capitalist's choice, however, we also control for the lead venture capitalist's reputation in order to factor in the two-sided (but likely deferred, staged) nature of the process.

We determine the lead venture capitalist in a given syndicate according to three criteria. When the first venture capital investment of a specific portfolio company is an individual one, the in-

vestor in that first round is the lead investor in the subsequent, syndicated round (Sorenson and Stuart, 2008). If the first venture capital investment in a specific portfolio company is done by a syndicate, we define the lead venture capitalist as the one with the largest investment sum in that round (Dai and Nahata, 2015; Wright and Lockett, 2003). If the lead investor cannot be identified by applying these first two criteria, we draw on the description

Table 6

Empirical analysis: The relative impact of partner reputation on the probability of cooperation in a syndicate based on low to medium reputation of lead investors.

	Low to medium lead investor reputation			
	International		Domestic	
	Model 1	Model 2	Model 3	Model 4
Dependent variable:	Cooperation	Cooperation	Cooperation	Cooperation
Independent variables and interactions:				
PARTNER_int_rep	0.0369*** (0.00567)		0.0368*** (0.00573)	
PARTNER_int_rep x PARTNER_local	−0.00295 (0.00602)		−0.00195 (0.00610)	
PARTNER_dom_rep		0.0481*** (0.00789)		0.0465*** (0.00786)
PARTNER_dom_rep x PARTNER_local		−0.0166** (0.00808)		−0.0144* (0.00807)
PARTNER_local	−0.392** (0.158)	−0.425*** (0.160)	−0.445*** (0.167)	−0.494*** (0.168)
Control variables:				
LEAD_PARTNER_cultdist4	0.0195 (0.0143)	0.0158 (0.0142)	0.0159 (0.0151)	0.0118 (0.0149)
LEAD_PARTNER_geodist	−0.000202*** (1.70e−05)	−0.000202*** (1.69e−05)	−0.000207*** (1.71e−05)	−0.000206*** (1.71e−05)
PARTNER_target_indepx5	0.661*** (0.0174)	0.662*** (0.0174)	0.657*** (0.0175)	0.658*** (0.0175)
Round_synsize	−0.000226 (0.00542)	−0.000117 (0.00546)	−0.00194 (0.00541)	−0.00191 (0.00544)
Target_Market_Change_t3	0.0873** (0.0408)	0.0784* (0.0416)	0.106** (0.0411)	0.0970** (0.0419)
Target_stage	−0.00380 (0.00687)	−0.00318 (0.00693)	−0.00336 (0.00689)	−0.00263 (0.00694)
Target_industry_FE	yes	yes	yes	yes
LEAD_nation_FE	yes	yes	yes	yes
Inv_year_FE	yes	yes	yes	yes
Constant	−2.456*** (0.159)	−2.493*** (0.161)	−2.408*** (0.166)	−2.427*** (0.167)
Observations	34,419	34,419	34,243	34,243
Pseudo R-squared	0.103	0.106	0.103	0.106

Note: In this table, we present results of binomial logistic regressions with the probability of cooperation between lead investor and co-investors in a syndicate as the dependent variable for subsamples of low to medium reputation of lead investors. Model 1 and Model 2 include only triads in which international reputation of the lead investor is low to medium, and Model 3 and Model 4 include only triads in which domestic reputation of the lead investor is low to medium. As independent variables, we use international and domestic reputations of syndicate partners and interactions terms thereof with the dummy variable PARTNER_local, which equals one when the syndicate partner is located in the same country as the portfolio company and zero otherwise. Heteroskedasticity robust standard errors are reported in parentheses, significance is marked by * at 10%, ** at 5%, and *** at 1%. Standard errors are clustered around portfolio firms. The interaction terms are displayed graphically in Fig. 3(a to d).

of the investors' preferred investment styles and mark the investor whose preferred investment style is "deal originator" as the lead. We exclude those rounds for which a lead investor cannot be determined from our analysis.

In order to clearly define a distinct scenario for internationality, we keep the lead venture capitalist's location relative to the portfolio company fixed. Namely, we restrict our dataset to lead venture capitalists investing in their own nations, with partners coming from either that country or abroad.

Our dataset is based on triads consisting of information on the lead venture capitalist, its syndicate partner, and the portfolio company in which they invest jointly. These triads form the unit of observation for our analysis.

The ThomsonONE database from which we built our dataset, contains only records of deals and co-investments that were entered into in practice. Potential syndicate partners that might have been considered during the decision-making process but with whom no cooperation materialized cannot be captured in the data. We therefore follow Sorenson and Stuart (2008) and match 10 control triads to each realized one. These control triads match the partner with whom the lead venture capitalists co-invested in reality insofar as they invested in the same primary industry in the same year and have demonstrated their willingness to invest in

the specific target country. That is to say, the matched partners invested in the same industry (i.e. as co-investors in a syndicate), in the same period (i.e. in the same year), and in the same target nation as the realized partners. Furthermore, the matched partners come from the same nation as the realized partners, where lead investors choose domestic partners, as we argue it is likely that the lead investor consciously decided whether to take on a foreign or domestic partner. Where lead investors choose a foreign partner, we matched other foreign partners based on their willingness to invest in the same year, industry, and target nation. In this way, our matched sample closely reflects the realized deals, as the proportions of purely domestic syndicates and multinational ones are exactly the same for realized and unrealized ties. Cases for which fewer than 10 matches could be identified were dropped.

Each control triad consists of the same lead venture capitalist and portfolio company as the respective realized case and the matched syndicate partner. Therefore, the control triads represent joint investments that could have materialized but did not. Altogether, our dataset consists of 46,101 triads, of which 4191 are realized cases and 41,910 are matched controls.

We summarize the national distribution of lead and partner venture capitalists as well as the distribution of deals among target nations in Tables 1.1 and 1.2. They show the distribution of

lead and partner venture capitalists' nationalities. The data in these tables differentiate among realized cases (Table 1.1) and matched cases (controls) in Table 1.2. As we matched 10 partners to each case, the numbers in Tables 1.1 and 1.2 add up to the number of triads in our full sample.

3.3. Variable description

3.3.1. Dependent variable: COOPERATION

The dependent variable COOPERATION is a binary variable that equals one for a triad that contains the partner selected to a venture capital syndicate and zero otherwise. In the case-control framework we apply for our analysis, COOPERATION therefore equals one for all realized cases and zero for all controls. This approach is consistent with a stream of literature on relationship formation in venture capital, including that of Bottazzi et al. (2016), Huyghebaert and Priem (2015), and Sorenson and Stuart (2008).

3.3.2. Major variable of interest: reputation

In order to comprehensively portray venture capital reputation, we resort to a number of different proxies. So far, no consensus on how to measure reputation in the venture capital industry has been established. Rindova et al. (2005) identify two dimensions of organizational reputation, namely the prominence of an actor among its stakeholders and the quality of the output. In the context of venture capital, this translates into three key components: the level of investment activity, the survival of the fund and exit performance.

3.3.3. Reputation components: investment activity

The prominence and visibility of a venture capitalist are strongly related to its level of activity, not least because a high activity level corresponds to being in contact with many actors (Fund et al., 2008; Lee et al., 2011; Rindova et al., 2007). We analyze three measures characterizing the particular partner's level of activity. These comprise the total dollar amount of equity invested over the last ten-year period as well as the total number of international and domestic portfolio companies invested in.

3.3.4. Reputation components: survival and fund management

A venture capitalist's quality, as perceived among investors, is demonstrated by its survival in the market and its continued ability to raise funds. Survival in the market is synonymous with the age of a specific venture capitalist, which also indicates a proven track record and the length of time it has been able to continue operations by raising funds successfully. This ability is also reflected in the average number of funds managed (Lee et al., 2011). We use proxies, venture capital firm age, and the average number of funds managed in our empirical analysis. Due to data restrictions with respect to the total dollar amount of funds under management, we are not able to include that dimension in our indices.

3.3.5. Reputation components: exit performance

Another major signal for the quality of a venture capitalist's output is the number of IPOs a venture capitalist has achieved for its portfolio companies. IPOs are viewed as the most profitable type of exit and therefore prove a venture capitalist's quality and guarantee publicity (Lee et al., 2011). Therefore, we use the number of IPOs as another component of reputation in our empirical analyses.

3.3.6. Reputation indices: international and domestic reputation

While many researchers have used one or a number of these single components as proxies for reputation (Gompers and Lerner, 1999a; Gompers, 1996; Lee et al., 2011; Lee and Wahal, 2004),

others argue that multi-item reputation indices decrease the effects of random errors and therefore come closer to the actual value of reputation (Boyd et al., 2005; Lee et al., 2011). We therefore construct two reputation indices and in addition focus on the individual components of venture capital reputation. The indices are constructed according to the methodology proposed by Lee et al. (2011). Accordingly, we take into consideration the respective total dollar amount invested over the last ten years, the number of start-ups invested in, the venture capital firm age, the ten-year average of the number of funds under management, and the number of portfolio companies taken public.

As we assume that local and international reputations are likely to differ, we construct two scores: one for international reputation and one for domestic reputation. We differentiate the two reputation indices according to one of the five measures in the reputation index, namely, the total number of start-ups invested in. For international reputation, we use the total number of start-ups invested in outside a venture capitalist's home market, and for domestic reputation, the number of investments within the home market. As we do not have access to more detailed data on equity invested, the number of funds raised, or the number of IPOs achieved in a particular country, the international and domestic reputation indices do not differ in these components. Despite this limitation, we use both proxies to take into account the international context and to detect nuances in venture capital syndication. Most likely, the number of domestic investments does not have the same impact on reputation on the international venture capital scene as it does in the local, domestic market. What counts more in the international scene are cross-border deals, as these add credibility to a venture capitalist's ability to successfully operate outside its home market and to add value to international investments.

To make the measures comparable, all of them were standardized by computing z-scores before aggregating them. The reputation scores were computed in rolling ten-year-windows, i.e. for each individual year for each venture capitalist in our sample; the information used to calculate the score for 2005 stems from each venture capitalist's operations in the previous ten years (or in the venture capitalists' total lifetime when they had been operating for less than ten years). This enabled us to capture fluctuations in reputation (Lee et al., 2011).

3.3.7. Moderator: location of a potential partner

We test whether the location of a potential partner influences the relevance of venture capital reputation to the probability of cooperation in a syndicate. Specifically, we analyze whether reputation is more or less important when the potential co-investor is located in the same nation as the target company rather than abroad. In addition to analyzing the relative impact of the aggregated reputation scores, we also test the impact of each of the individual components of our reputation measures.

3.3.8. Further independent variables and controls

In our framework, we are confronted with three different dimensions, each of which contains particular factors that are potentially relevant to the lead venture capitalist's choice of syndicate partner. These include the quality of the match between the lead and its syndicate partner, the potential partner's qualifications and fit to the designated target, and the suitability and fit of the lead venture capitalist and its investment target. Including all three dimensions in our analysis gives rise to empirical problems due to their correlation. We therefore chose to restrict the model to the level of fit of the potential partner to the lead itself and the target, assuming that the partner's suitability on both levels is what matters most for the lead's selection decision.

For the syndicate level, we control for characteristics that indicate the level of fit between the lead and partner venture capital-

ists. As the major focus of our analysis is on international relations in venture capital deals, we follow Dai and Nahata (2015) and include a measure for the difference in culture between the nations in which the lead and partner are headquartered. We also control for the geographic distance between lead and partner venture capitalists by applying the geodetic distance between their respective headquarters in miles.

In terms of the quality of fit between the partner and the designated target, we control for the partner's level of expertise in the target industry in terms of the industry in which that partner made the most investments over the previous five years.

We further control for the state of the venture capital market in the target nation (Plagmann and Lutz, 2019), for the portfolio company's industry and stage, and for each respective investment year by including annual dummies. A complete set of country fixed effects for lead venture capitalists controls for nation-specific effects, such as the legal and institutional environment, as well as the state of the venture capital markets (Bottazzi et al., 2016; La Porta et al., 1998). The full set of variables used in all our models is explained in Table 2, and descriptive statistics can be found in Table 3.

4. Results and discussion

Our empirical analysis proceeds according to the order of our hypotheses. We use logit models with robust standard errors clustered around the portfolio company to estimate the probability of a potential partner being selected to the syndicate by the lead venture capitalist. The correlation matrix and variance inflation factors for the variables of key interest are reported in Table 4.

4.1. Relative impact of reputation on the lead venture capitalist's syndicate partner choice

Our sample contains deals in which the lead venture capitalist invests locally and includes potential partners from the same nation as well as from abroad. Therefore, we run the same model using the partners' international (Table 5, Model 1) and domestic reputations (Table 5, Model 2). We further control for the level of fit among the lead and partner investors at the syndicate level, as well as between the respective investors and the designated investment target.

We find that the partner's aggregated reputation scores (both international and domestic) have a positive impact on the probability of cooperation for both domestic and international partners. To get a better grasp on the economic effect, we analyze the percentage change in the probability of being selected for the syndicate by the lead investor based on a change ranging from zero to the mean of each aggregated reputation score +3 standard deviations. We decided to show this asymmetric range, as the reputation scores turn negative at values lower than mean -0.5 standard deviations. The illustrations are provided in Fig. 1.

For both local and cross-border partner venture capitalists, we find that the relative probability of being selected increases with higher reputation. This is true for both aggregated reputation scores: international as well as domestic. Reputation, however, has a larger impact on the lead venture capitalist's selection process if cross-border rather than local partners are concerned, which is in line with our hypothesis. Looking at local and international partners endowed with international (domestic) reputation at the mean, we see that the probability of being selected into a syndicate is 8.66% (8.55%) for local partners, and 10.87% (12.47%) for international ones. The difference becomes more pronounced for high values of partners' reputations. For potential partners with international (domestic) reputation values at the mean +2 standard deviations, the probability of being selected is 14.44% (14.70%)

for local partners and 19.91% (28.14%) for international ones (see Fig. 1a and b).

4.2. Disentangling the relative impact of the components of reputation for the lead venture capitalist's syndicate partner choice

In order to generate further insight into the relative impact of the various components of our reputation measures, we estimate the same model again, focusing on one reputation component at a time. The results of these models (Models 3–8) are presented in Table 5 and depicted graphically in Fig. 2.

In Model 3 (see Fig. 2a), we focus on the total dollar amounts of equity invested by potential partners; the results are similar to those in Models 1 and 2. The lead venture capitalist places more emphasis on a potential partner's total equity invested, and therefore on its investment activity over the previous ten years, when selecting an international partner than when choosing a partner locally. Looking at the mean value of this reputation proxy, the probability of being selected into the syndicate based on the equity invested is 8.58% for local partners and 12.05% for international ones. One might interpret the total dollar amount of equity invested to be a proxy for a good track record (as that helps raise funds, which can then be invested). When partners are invited to a syndicate to secure access to foreign capital markets (to which domestic partners are unlikely to have access), it seems quite reasonable that the partner's track record of raising capital over the past ten years matters more for these partners than for domestic ones (Gu and Lu, 2014). Furthermore, the total dollar amount of equity invested might be a proxy for a possible abundance of capital. This may point to another reason for choosing international partners, namely a potential need for additional capital rather than a need for additional resources (Hochberg et al., 2015; Manigart et al., 2002).

In Model 4, we focus on the number of portfolio companies invested in internationally (outside the partner's home country) over the last ten years. Based on the graphical illustration in Fig. 2b, we find that this reputation proxy has a slightly larger impact on the lead venture capitalist's selection decision for potential foreign partners than for potential domestic partners. Where foreign partners are chosen as co-investors for an investment in the lead venture capitalist's home nation (which is the case in our scenario by definition), they need international investment expertise. For domestic partners who become co-investors in a local portfolio company, international expertise is not a necessity; this might explain why the number of international investments does not matter as much for these local partners as for international ones. Unlike the pattern in the previous models, we find that the spread of the impact of this reputation proxy on the probability of cooperation narrows where we get closer to very high values at the mean +3 standard deviations. That might stem from the fact that an outstanding number of international investments is a signal of remarkable quality for international and domestic partners alike. So while international investment experience might not be a necessary requirement for local partners, an exceptionally high value of this proxy has an impact on the probability of a local partner being selected for a syndicate similar to that of international partners. The interaction term in Model 4 is, however, positive and not significant.

In Model 5 (see Fig. 2c), we analyze the impact of potential partners' investments in domestic portfolio companies over the last ten years. This proxy follows the same general pattern, as its impact on the probability of being selected into a syndicate is larger for international rather than local partners. Unlike the number of international investments, however, we find that the difference actually increases at higher values. That is, at the mean value of 23.18 domestic investments over the previous ten years, the probability

Table 7

Empirical analysis: The relative impact of partner reputation on the probability of cooperation in a syndicate based on high reputation of lead investors.

	High lead investor reputation			
	International		Domestic	
	Model 1	Model 2	Model 3	Model 4
Dependent variable:	Cooperation	Cooperation	Cooperation	Cooperation
Independent variables and interactions:				
PARTNER_int_rep	0.0546*** (0.0105)		0.0550*** (0.0106)	
PARTNER_int_rep x PARTNER_local	−0.0174 (0.0111)		−0.0204* (0.0110)	
PARTNER_dom_rep		0.0653*** (0.00989)		0.0679*** (0.0103)
PARTNER_dom_rep x PARTNER_local		−0.0307*** (0.0104)		−0.0351*** (0.0107)
PARTNER_local	0.177 (0.235)	0.0709 (0.243)	0.261 (0.208)	0.187 (0.219)
Control variables:				
LEAD_PARTNER_cultdist4	0.0286 (0.0210)	0.0228 (0.0208)	0.0357* (0.0190)	0.0308 (0.0189)
LEAD_PARTNER_geodist	−0.000110*** (2.17e−05)	−0.000109*** (2.16e−05)	−0.000102*** (2.14e−05)	−0.000102*** (2.14e−05)
PARTNER_target_indepx5	0.725*** (0.0309)	0.724*** (0.0310)	0.740*** (0.0303)	0.738*** (0.0304)
Round_synsize	−0.00804 (0.00967)	−0.00637 (0.00981)	0.00102 (0.00936)	0.00260 (0.00959)
Target_Market_Change_t3	0.0211 (0.0741)	0.00464 (0.0787)	−0.0296 (0.0734)	−0.0468 (0.0803)
Target_stage	0.000755 (0.0131)	0.000976 (0.0133)	−0.00247 (0.0130)	−0.00219 (0.0131)
Target_industry_FE	yes	yes	yes	yes
LEAD_nation_FE	yes	yes	yes	yes
Inv_year_FE	yes	yes	yes	yes
Constant	−3.264*** (0.233)	−3.235*** (0.237)	−3.317*** (0.209)	−3.323*** (0.216)
Observations	11,682	11,682	11,858	11,858
Pseudo R-squared	0.119	0.123	0.120	0.123

Note: In this table, we present results of binomial logistic regressions with the probability of cooperation between lead investor and co-investors in a syndicate as the dependent variable for subsamples of high reputation of lead investors. Model 1 and Model 2 include only triads in which international reputation of the lead investor is high, and Model 3 and Model 4 include only triads in which domestic reputation of the lead investor is high. As independent variables, we use international and domestic reputation of syndicate partners and interactions terms thereof with the dummy variable PARTNER_local, which equals one when the syndicate partner is located in the same country as the portfolio company and zero otherwise. Heteroskedasticity robust standard errors are reported in parentheses, significance is marked by * at 10%, ** at 5%, and *** at 1%. Standard errors are clustered around portfolio firms. The interaction terms are displayed graphically in Fig. 4(a to d).

of being selected for a local partner is 8.28% and 15.10% for an international one. At the mean +2 standard deviations, the probability of being selected is 15.51% for a local partner and 34.73% for an international one. This might indicate that lead investors turn abroad in their search for syndicate partners if they are looking for particular skills and resources to add value. If this is the case, experience in terms of the total number of investments made (possibly specialized in terms of the portfolio companies' industry or stage) would have to compensate for the additional risk involved by choosing a dissimilar partner.

In Model 6 (Fig. 2d), we focus on the potential partners' age and find a pattern largely similar to previous models. The lead venture capitalist places more emphasis on a potential partner's age, and therefore on its survivability, which signals a good track record and the ability to continuously raise funds, when selecting an international partner rather than a domestic one (Lee et al., 2011). Domestically, where lead venture capitalists might be able to interact with syndicate partners more closely and have access to more information regarding potential partners' investment managers and their expertise through their network, age itself might not be as important a criterion as for international partners.

Model 7 (Fig. 2e), in which we analyze the relative impact of the average number of managed funds on the lead investor's selection decision, also follows the pattern we have observed so far: this measure has a higher impact for international partners than for

domestic ones. Analyzing the relative impact of companies taken public in Model 8 (see Fig. 2f), we find that this measure is more important to the lead investor where international partners are concerned, with the difference increasing as the number of IPOs rises above the mean value. This lends support to the argument that local lead investors choose international partners for a particular reason, and one major reason might be their expertise in ex-its (possibly in the foreign partner's domestic market). For venture capitalists in China, Gu and Lu (2014) find that they are rather dependent on international syndicate partners, as IPOs usually take place overseas. If an international IPO is the strategic reason for cross-border syndication, a track record of IPOs would, of course, matter in the lead investor's partner selection.

Our results are consistent across all proxies for venture capital reputation, including the two overall reputation indices as well as the single components, and we can therefore confirm Hypothesis 1.

4.3. The relative impact of partner reputation on lead venture capitalists' foreign syndicate partner choice – boundary conditions based on lead venture capitalists' reputation

To analyze the results obtained above based on the lead venture capitalists' own levels of reputation, we use several subsamples of our data. These are split according to the lead venture capitalists' level of reputation. As our analysis of the individual

Table 8

Robustness check: The relative impact of reputation and various components on the probability of cooperation in a syndicate in a non-US-sample.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Dependent variable:	Cooperation	Cooperation	Cooperation	Cooperation	Cooperation	Cooperation	Cooperation	Cooperation
Independent variables and interactions:								
PARTNER_int_rep	0.0645*** (0.00732)							
PARTNER_int_rep x PARTNER_local	−0.0533*** (0.0111)							
PARTNER_dom_rep		0.0674*** (0.00817)						
PARTNER_dom_rep x PARTNER_local		−0.0443*** (0.0132)						
PARTNER_equityinv			0.000580*** (8.60e-05)					
PARTNER_equityinv x PARTNER_local			−0.000399** (0.000176)					
PARTNER_int_pfc				0.0255*** (0.00424)				
PARTNER_int_pfc x PARTNER_local				−0.0261*** (0.00489)				
PARTNER_dom_pfc					0.0215*** (0.00291)			
PARTNER_dom_pfc x PARTNER_local					−0.0166*** (0.00332)			
PARTNER_age						0.0501*** (0.00987)		
PARTNER_age x PARTNER_local						−0.0487*** (0.0151)		
PARTNER_fundsmgd							0.508*** (0.0622)	
PARTNER_fundsmgd x PARTNER_local							−0.404*** (0.0805)	
PARTNER_ipos								0.0655*** (0.0112)
PARTNER_ipos x PARTNER_local								−0.00914 (0.0230)
PARTNER_local	0.576*** (0.201)	0.499** (0.220)	0.425** (0.200)	0.405** (0.171)	0.279 (0.204)	0.634*** (0.236)	0.708*** (0.221)	0.115 (0.178)
Control variables:								
LEAD_PARTNER_cultdist4	0.0271 (0.0176)	0.0328* (0.0177)	0.0402** (0.0183)	0.0295* (0.0166)	0.0267 (0.0174)	0.0379** (0.0161)	0.0307* (0.0174)	0.0247 (0.0162)
LEAD_PARTNER_geodist	−0.000120*** (4.24e-05)	−0.000140*** (4.23e-05)	−0.000127*** (4.30e-05)	−7.53e-05* (3.88e-05)	−0.000141*** (4.10e-05)	−0.000109*** (3.81e-05)	−0.000126*** (4.26e-05)	−0.000108*** (3.91e-05)
PARTNER_target_indepx5	0.575*** (0.0471)	0.576*** (0.0468)	0.563*** (0.0470)	0.570*** (0.0473)	0.579*** (0.0472)	0.568*** (0.0462)	0.584*** (0.0471)	0.569*** (0.0467)
LEAD_int_reputation	0.00211 (0.00178)	0.00249 (0.00175)	0.000752 (0.00176)	0.00118 (0.00176)	0.00324* (0.00165)	0.00343* (0.00198)	0.00292 (0.00180)	0.00261 (0.00163)
Round_syssize	−0.0141 (0.0193)	−0.0205 (0.0189)	−0.0213 (0.0190)	−0.0168 (0.0183)	−0.0266 (0.0194)	−0.0330* (0.0178)	−0.0192 (0.0193)	−0.0235 (0.0185)
Target_Market_Change_t3	−0.0143 (0.0300)	0.00338 (0.0314)	−0.00594 (0.0310)	−0.0288 (0.0252)	0.00181 (0.0319)	−0.00715 (0.0273)	−0.00912 (0.0303)	−0.00480 (0.0312)
Target_stage	0.0112 (0.0191)	0.0159 (0.0183)	0.0132 (0.0176)	−0.00202 (0.0184)	0.0201 (0.0182)	0.00967 (0.0163)	0.0155 (0.0193)	0.0146 (0.0170)
Target_industry_FE	yes	yes	yes	yes	yes	yes	yes	yes
LEAD_nation_FE	yes	yes	yes	yes	yes	yes	yes	yes
Inv_year_FE	yes	yes	yes	yes	yes	yes	yes	yes
Constant	−3.778*** (0.233)	−3.807*** (0.244)	−3.586*** (0.236)	−3.496*** (0.215)	−3.502*** (0.236)	−3.739*** (0.245)	−3.988*** (0.245)	−3.314*** (0.217)
Observations	4972	4972	4972	4972	4972	4972	4972	4972
Pseudo R-squared	0.0844	0.0849	0.0824	0.0747	0.0845	0.0681	0.0866	0.0744

Note: In this table, we present results of a robustness check in a sample excluding deals in the US-market. We show results of a binomial logistic regressions with the probability of cooperation between lead investor and co-investors in a syndicate as the dependent variable. As independent variables, we use various metric proxies for syndicate partner reputation and interactions terms thereof with the dummy variable PARTNER_local, which equals one when the syndicate partner is located in the same country as the portfolio company and zero otherwise. Heteroskedasticity robust standard errors are reported in parentheses, significance is marked by * at 10%, ** at 5%, and *** at 1%. Standard errors are clustered around portfolio firms. The interaction terms of Model 1 and Model 2 are displayed graphically in Fig. 5(a/b), and the interaction terms of Models 3 to 8 are displayed in Fig. 6(a to f).

reputation components has proven rather consistent with the results obtained from the models using the aggregated reputation scores, we focus on these two scores again for this analysis. For each of the aggregated reputation scores (international and domestic), we divide the data into two subsamples, one for low-to-medium levels of lead venture capitalists' reputations (i.e. reputation levels of up to the 75th percentile) and one for lead venture capitalists' reputations above the 75th percentile. Overall, there-

fore, we obtain four subsamples, two for each of the aggregated reputation scores.

4.3.1. Lead venture capitalists with low-to-medium reputation and the impact of partner reputation on their syndicate partner choice

We start by modeling the relative impact of partner reputation on the lead venture capitalists' selection process for situations in which the lead venture capitalist itself commands a low-

to-medium reputation. The results are depicted in Table 6, Models 1–4 and illustrated graphically in Fig. 3.

When focusing on low-to-medium levels of international reputation on the lead venture capitalist side, we assume that these leads have low bargaining power in their search for international partners because they are the ones approaching partners abroad (probably due to a specific need for endowments they cannot obtain among partners domestically) and because syndicating with them might not be particularly appealing for foreign partners. So, as stated in Hypothesis 2, we assume these lead venture capitalists are in the position of being beggars rather than choosers. Examining the impact of potential partners' international reputations based on lead venture capitalists who themselves command low-to-medium levels of international reputation, we find a general pattern similar to the one observed in our previous models (see Table 6, Model 1, and Fig. 3a). So once more, partners' international reputations have a greater impact on the lead venture capitalists' selection decisions if these partners are international than when they are from the same nation. The same is true when we replace partners' international reputation score with their aggregated reputation in their home market in our model (see Table 6, Model 2 and Fig. 3b). This proxy is also of higher importance in the selection decision for foreign partners than for local ones. The difference among the impact of local versus international partners' reputation is rather stable across the range of values for their international reputation. For potential partners' domestic reputation, however, we observe a spread at above-average values of reputation, where international partners commanding the same reputation as domestic ones are favored even more. That lends support to the argument that international partners are selected due to certain expertise in their specific home markets, for example when planning an IPO in that nation, which domestic partners have only in their respective home markets.

As our dataset comprises only deals made by lead venture capitalists in their home markets, their reputations in these markets might be linked to their bargaining position when asking foreign partners to join the syndicate. High domestic reputation in the market in which the identified deal takes place might give lead venture capitalists credibility and therefore bargaining power when trying to attract investors from abroad, while low domestic reputation will probably not be helpful in attracting potential partners from abroad. Running the same two models in our next subsample, which consists of deals with lead venture capitalists commanding low and medium reputations in their home markets, we obtain results similar to the previous case. Both the potential partners' international reputation scores (see Table 6, Model 3 and Fig. 3c) and potential partners' domestic reputation scores (see Table 6, Model 4 and Fig. 3d) have a higher impact on the lead venture capitalist's partner choice when these partners are headquartered abroad than when they come from the same nation as the lead itself. The spread between both scores is rather stable across the values of potential partners' international reputation, while it widens for values above the mean, namely above-average values of domestic reputation. Once more, this supports the argument that international partners are chosen for their specific expertise in their respective home markets.

Based on these results, we therefore conclude that the lead venture capitalists' having a low-to-medium reputation themselves does not lead to a substantial deviation from the general pattern observed in our previous analysis.

4.3.2. Lead venture capitalists with high reputations and the impact of partner reputations on their syndicate partner choice

We re-run our models to assess the relative impact of partner reputations when the lead venture capitalists themselves com-

mand high levels of reputation. The results are depicted in Table 7, Models 1–4 and illustrated graphically in Fig. 4.

As stated previously, we assume that the lead venture capitalists' own reputations influence their bargaining power and the attractiveness of their invitation to join their syndicate. High reputation on the international scene should therefore put them in the best possible position to attract syndicate partners with the necessary endowments that may not be found in domestic partners.

The relative importance of international and domestic partner reputations deviates from the previous pattern in both subsamples of lead investors with high reputation. Regarding the international and domestic reputation scores of co-investors, we find a steeper slope for potential international syndicate partners than for domestic ones. The curves showing the probability values for international and local partners dependent on their reputations intersect shortly above the mean value in both subsamples for lead venture capitalists with a high reputation (high international reputation is shown in Fig. 4a and b; high domestic reputation is shown in Fig. 4c and d). The slope of the probability of cooperation curve with cross-border co-investors is steeper than that of the curves concerning domestic co-investors. Hence, high-reputation lead investors do not generally demand a higher reputation premium when selecting cross-border syndicate partners. Instead, they are more likely to cooperate with cross-border co-investors when their international reputation is high, i.e. above the intersection point around the mean value; they are also more likely to cooperate with domestic rather than international partners below the intersection point. It seems that highly reputable lead investors choose international syndicate partners more rigorously in terms of their international reputation compared to that of local syndicate partners. Due to the risks associated with international syndicates as well as specific value-adding requirements for international syndicate partners, highly reputable lead investors are particularly picky in their selection process, and probably can afford to be, as their high reputation renders them choosers rather than beggars.

Overall, we reject Hypothesis 2, as we do not find evidence of highly reputable lead venture capitalists focusing on higher reputation premiums in order to offset the additional dissimilarity international partners bring to syndicates. Instead, they seem to follow a more intricate decision model when selecting their syndicate partners. The high international reputation of cross-border syndicate partners is strongly valued by highly reputable lead investors.

4.4. Robustness check: excluding the US market

Almost 90% of the observations in our sample refer to investments in portfolio companies located in the US. The US venture capital market is the largest market globally and likely to be the most developed one as well. Based on total capital allocated globally over the last 20 years, 59% of global venture capital investments went to the US market (Bottazzi and Da Rin, 2002). Due to the unique status of the US market within the global venture capital sphere, the composition of our sample does not deviate much from reality. The unique nature of the US-market, however, might render conditions for investment, syndication, and partner selection different from conditions in other nations, and the results of our analysis so far might be driven by the circumstances in the US market. We therefore perform a robustness check in which we re-run all our models in a sample while excluding all observations with portfolio companies in the US. The results of our basic models in which we analyze the aggregate reputation scores as well as single reputation measures are provided in Table 8 and illustrated in Figs. 5 and 6.

Table 8 shows that results are largely robust. The general pattern of the results obtained in the sample including the US market—namely, that international, rather than local partners are

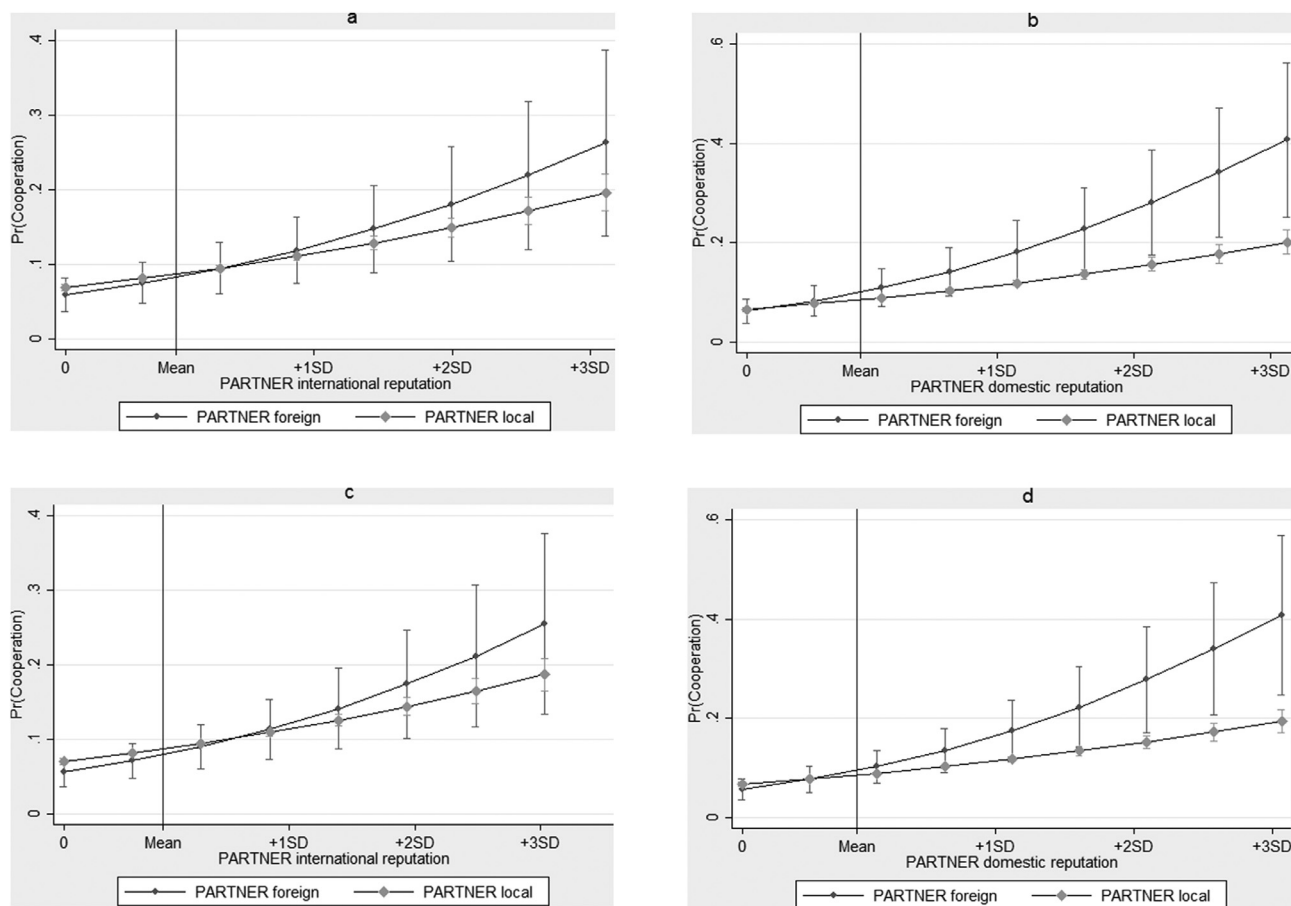


Fig. 4. Impact of potential local vs. foreign partners' reputations on the probability of cooperation for subsamples of high lead investor reputation.

Note: Figs. 4(a) and 4(b) are based on a subsample of triads including only lead investors with high international reputation. Figs. 4(c) and 4(d) are based on a subsample of triads including only lead investors with high domestic reputation. The figures present calculated predictions of Model 1 to Model 4 (Table 7). We show predictive margins with 95% confidence intervals for foreign and local partners, respectively, depending on the values of PARTNER_international_reputation (Fig. 4(a) and Fig. 4(c), and PARTNER_domestic_reputation (Fig. 4(b) and Fig. 4(d)).

chosen at the same level of reputation—is sustained. Models 1–7 (see Fig. 5a and b and Fig. 6a–e), show a peculiar pattern, however. For all of these models, the slope of the graph corresponding to international partners is much steeper than for local ones, with both intersecting at or slightly above the mean value. Local lead venture capitalists in markets outside the US apparently choose local rather than international partners in cases of low-to-medium reputation and choose international partners at high levels of reputation. This might stem from the fact that the markets outside the US are different in terms of regulation, culture, and institutions, and therefore pose a more difficult environment for foreign venture capitalists to operate in than does the US market and are therefore easier for foreigners to operate in. Bringing in foreign partners for investments in the more peculiar, smaller markets outside the US might simply not make sense if these foreign partners have low or average reputation, as the additional value they bring in might not compensate for their unfamiliarity with local circumstances. Highly reputable international partners, on the other hand, might have such valuable expertise that the liability of their foreignness, as well as the introduced dissimilarity in the syndicate, is offset.

Turning to Table 8, Model 8 (see also Fig. 6f), in which we test the partners' number of previous IPOs as a proxy for reputation, deviates even further from previous results, with foreign partners' number of IPOs having a slightly lower impact on the selection decision than local partner' number of IPOs. Lead venture capitalists outside the US seem to place slightly less importance on potential cross-border co-investors' IPO-track record compared to the

IPO experience of local co-investors.² This might be due to the superior role of US public equity markets compared to other, smaller countries in the sample. While the US offers attractive IPO opportunities for the portfolio companies of venture capitalists, in other markets trade sales might be the more relevant exit channel. IPO experience might therefore be valued less by lead investors located outside of the US.

4.5. Limitations and future research

Our research question is restricted to analyzing lead venture capitalists' selection of syndicate partners. The lead investors usually approach potential partners with an offer to join syndicates, as the syndicate's compilation is part of their conventional role. Syndicate formation is not a one-sided process, however, but rather a deferred, two-stage one (Fox et al., 2012; Wright and Lockett, 2003). After being approached by a lead venture capitalist, partners also evaluate the lead's proposal in order to decide whether to join the syndicate (Gu and Lu, 2014; Meuleman et al., 2009; Wright and Lockett, 2003). We touch upon this aspect by factoring in the lead venture capitalists' own reputations, and thereby one aspect of their respective offers' attractiveness, into our analysis. We do not, however, include the lead venture capitalists' offers that were rejected as we do with partners that have not been

² The results of the analysis of boundary conditions in the non-US-sample are available from the authors on request.

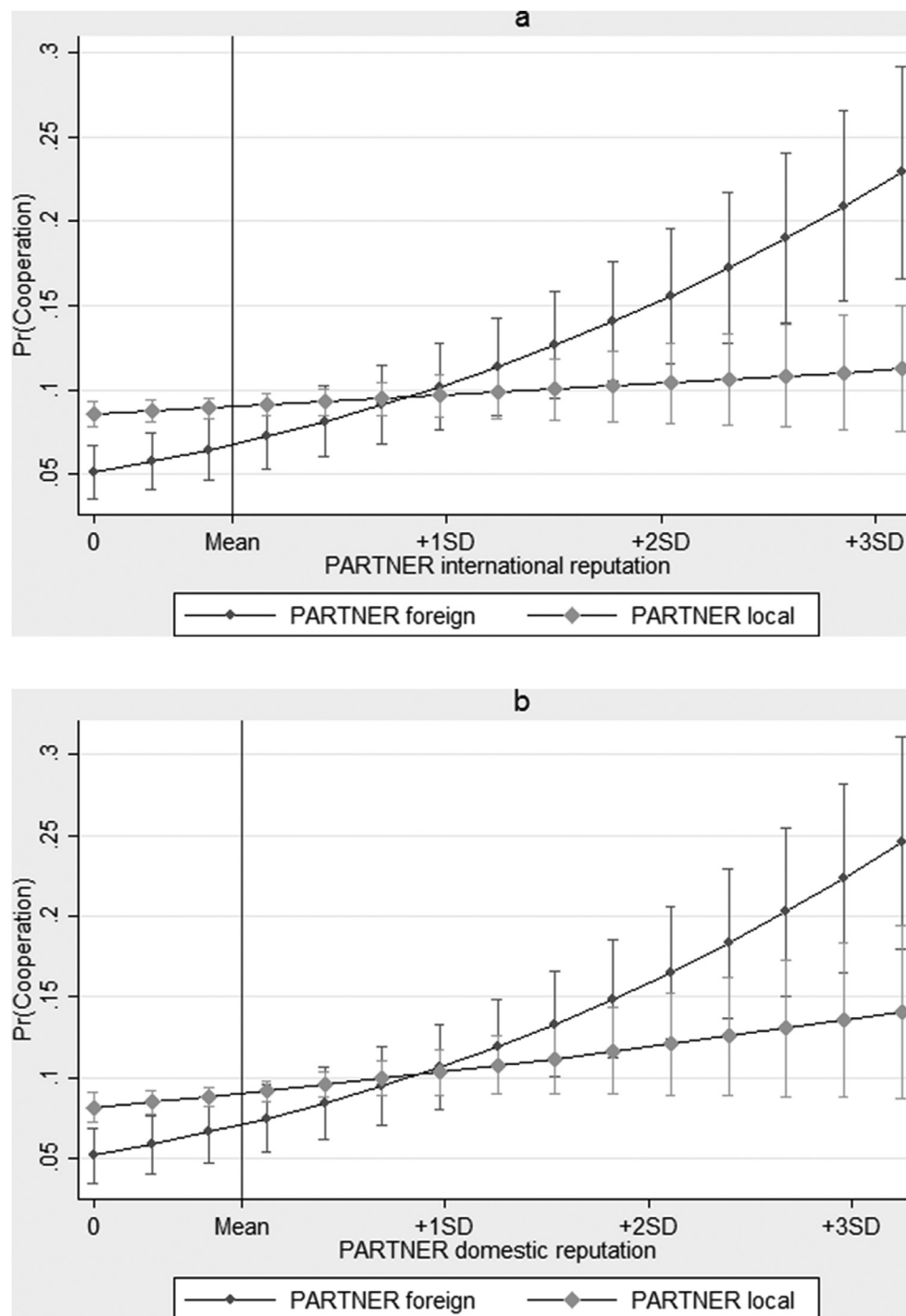


Fig. 5. Robustness check: impact of potential local vs. foreign partners' international and domestic reputations on the probability of cooperation in a sample excluding investments in the US-market.

Note: Figs. 5(a) and 5(b) present the calculated predictions of Model 1 and Model 2 (Table 8). We show predictive margins with 95% confidence intervals for foreign and local partners, respectively, depending on the values of PARTNER_international_reputation (Fig. 5(a), and PARTNER_domestic_reputation (Fig. 5(b)).

selected. Future research might analyze both decision-making processes in parallel to provide further insights into the potential partners' bargaining power. Fox et al. (2012) provide a possible framework for such an analysis.

Furthermore, we need to rely on observed, realized partnerships and construct the pool of potential partners artificially. A qualitative study would add knowledge regarding the relative importance of various partner characteristics to their selection into syndicates. In-depth interviews would allow for the identification

of partners that were considered during the decision-making process but not invited to the syndicate. Moreover, our analysis is restricted to modeling the lead venture capitalist's choice of each partner individually. Therefore, we fail to account for the potential impact of the resources, nationality, and reputation with which peers that are already part of the syndicate are endowed.

Next to reputation, bilateral trust is another mechanism used by venture capitalists to deal with the risk of syndicating with international partners previously unknown to them or mitigate

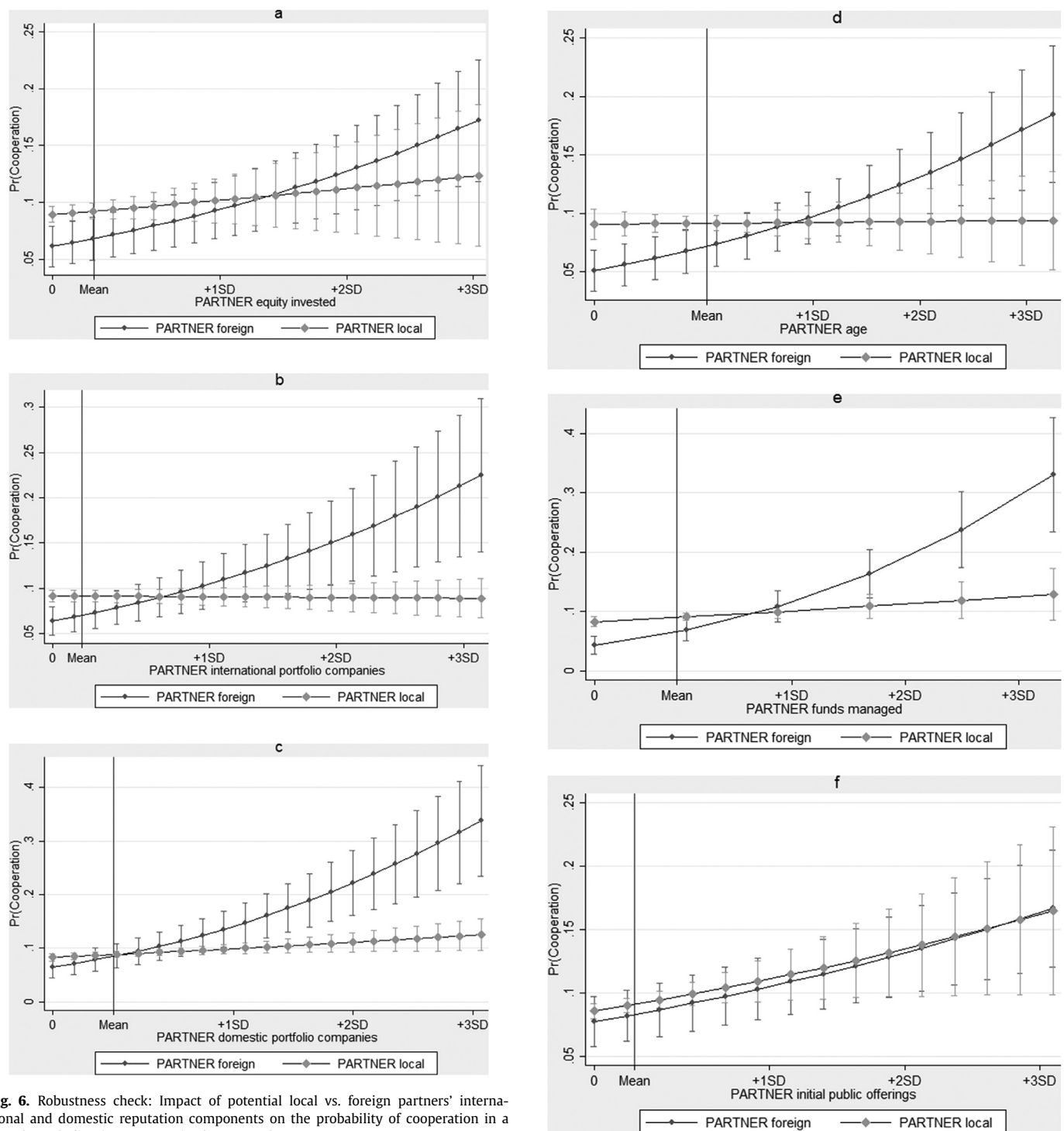


Fig. 6. Robustness check: Impact of potential local vs. foreign partners' international and domestic reputation components on the probability of cooperation in a sample excluding investments in the US-market.

Note: Figs. 6(a) to 6(f) present the calculated predictions of Model 3 to Model 8 (Table 8). We show predictive margins with 95% confidence intervals for foreign and local partners, respectively, depending on the values of potential partners' 6(a) equity invested, 6(b) number of investments in international portfolio companies, 6(c) number of investments in domestic portfolio companies, 6(d) age, 6(e) number of funds managed and 6(f) total number of IPOs.

the risks when choosing portfolio companies in foreign countries (Bottazzi et al., 2016; Plagmann and Lutz, 2019). Unfortunately, the measure of bilateral trust used in these studies is only available for a sample of 15 European nations, and the most recent collection by the European Commission was in 1997 (EuropeanCommission, 2012). To the best of our knowledge, no updated measure of bilateral trust for a global sample of countries is

Fig. 6. Continued

available, but it would be of great interest to review the relative impact of reputation on the lead investor's syndicate partner selection while considering trust.

Finally, we are unable to analyze the impact of international syndicates on performance of the respective deals. Previous studies have shed light on the fact that syndicates among foreign and domestic venture capitalists often have a positive impact on performance (Chemmanur et al., 2016; Cumming et al., 2016; Gu and Lu, 2014; Pruthi et al., 2003). Matching performance data to an analysis like ours would be of interest for future research in order to

test whether the selection mechanisms we have detected actually translate into tangible investment outcomes.

5. Conclusion

In this article, we investigate the relative impact of reputation on the lead venture capitalist's choice of syndicate partners. We focus on one particular scenario of internationality in syndicated venture capital investments, namely lead venture capitalists investing domestically with either local or international syndicate partners. Based on previous results on the importance of reputation for venture capital syndication (Atanasov et al., 2012; Gu and Lu, 2014; Zhelyazkov and Gulati, 2015), we analyze the moderating impact of internationality introduced by foreign syndicate partners. Furthermore, an analysis of boundary conditions based on the lead venture capitalists' own reputations and the potential bargaining power associated with them shows that high lead reputation is related to an intricate decision model.

This study contributes to the literature in several ways. First, it adds to the literature on reputation in the venture capital industry (Atanasov et al., 2012; Gu and Lu, 2014) and on international venture capital syndication (Chemmanur et al., 2016; Dai and Nahata, 2015; Mäkelä and Maula, 2006, 2008; Tykvová and Schertler, 2014), as it provides further insight into the importance of reputation for syndicate partner selection in international settings. We find that for lead venture capitalists investing domestically, a partner's reputation (both on the international scene as well as in their home market) as well as all individual reputation components play a greater role in foreign partners' probability of selection into the syndicate than it does for potential partners located in the lead venture capitalists' home country.

Second, the study contributes to the literature on familiarity within the venture capital industry (Gompers et al., 2016; Sorenson and Stuart, 2001; Trapido, 2007). Cross-border syndicate partners introduce dissimilarity into a syndicate, thereby potentially increasing investment risk. We show that highly reputable venture capitalists deal with this unfamiliarity by choosing their international co-investors carefully. For them, the probability of cooperation in an international syndicate is particularly high when the cross-border partners possess high international reputations that mirror their own high reputations.

Our study is of interest to practitioners in the venture capital industry. We provide insights into the relative importance of reputation to venture capitalists who consider forming international syndicates. Less reputable venture capitalists can learn from their reputable counterparts in focusing on international investment experience when choosing cross-border syndicate partners. Overall, our results highlight the importance of selecting foreign co-investors that provide additional international skills unavailable in the local market.

References

- Aizenman, J., Kendall, J., 2012. The internationalization of venture capital. *J. Econ. Stud.* 39, 488–511.
- Atanasov, V., Ivanov, V., Litvak, K., 2012. Does reputation limit opportunistic behavior in the VC industry? Evidence from litigation against VCs. *J. Finance* 67, 2215–2246.
- Bertoni, F., Groh, A.P., 2014. Cross-border investments and venture capital exits in Europe. *Corp. Govern.* 22, 84–99.
- Bottazzi, L., Da Rin, M., 2002. Venture capital in Europe and the financing of innovative companies. *Econ. policy* 17, 229–270.
- Bottazzi, L., Da Rin, M., Hellmann, T., 2016. The Importance of trust for investment: Evidence from venture capital. *Rev. Financ. Studies* 29, 2283–2318.
- Boyd, B.K., Gove, S., Hitt, M.A., 2005. Construct measurement in strategic management research: Illusion or reality? *Strategic Manage. J.* 26, 239–257.
- Brander, J.A., Amit, R., Antweiler, W., 2002. Venture-capital syndication: Improved venture selection vs. the value-added hypothesis. *J. Econ. Manage. Strategy* 11, 423–452.
- Bruton, G.D., Ahlstrom, D., 2003. An institutional view of China's venture capital industry: Explaining the differences between China and the West. *J. Bus. Venturing* 18, 233–259.
- Chemmanur, T.J., Hull, T.J., Krishnan, K., 2016. Do local and international venture capitalists play well together? The complementarity of local and international venture capitalists. *J. Bus. Venturing* 31, 573–594.
- Chen, H., Gompers, P., Kovner, A., Lerner, J., 2010. Buy local? The geography of venture capital. *J. Urban Econ.* 67, 90–102.
- Cumming, D., Dai, N., 2010. Local bias in venture capital investments. *J. Empirical Finance* 17, 362–380.
- Cumming, D., Dai, N., 2013. Why do entrepreneurs switch lead venture capitalists? *Entrepreneurship Theory Pract.* 37, 999–1017.
- Cumming, D., Knill, A., Syvud, K., 2016. Do international investors enhance private firm value? Evidence from venture capital. *J. Int. Bus. Studies* 47, 347–373.
- Cumming, D., Schmidt, D., Walz, U., 2010. Legality and venture capital governance around the world. *J. Bus. Venturing* 25, 54–72.
- Cumming, D.J., 2006. The determinants of venture capital portfolio size: Empirical evidence. *J. Bus.* 79, 1083–1126.
- Dai, N., Jo, H., Kassicheh, S., 2012. Cross-border venture capital investments in Asia: Selection and exit performance. *J. Bus. Venturing* 27, 666–684.
- Dai, N., Nahata, R., 2015. Cultural differences and cross-border venture capital syndication. *J. Int. Bus. Studies* 47, 140–169.
- Demiroglu, C., James, C.M., 2010. The role of private equity group reputation in LBO financing. *J. Finan. Econ.* 96, 306–330.
- Devigne, D., Manigart, S., Wright, M., 2016. Escalation of commitment in venture capital decision making: Differentiating between domestic and international investors. *J. Bus. Venturing* 31, 253–271.
- Devigne, D., Vanacker, T., Manigart, S., Paeleman, I., 2013. The role of domestic and cross-border venture capital investors in the growth of portfolio companies. *Small Bus. Econ.* 40, 553–573.
- Espenlaub, S., Khurshed, A., Mohamed, A., 2015. Venture capital exits in domestic and cross-border investments. *J. Banking Finance* 53, 215–232.
- European Commission, 2012. Eurobarometer 46.0 (Oct–Nov 1996), "Version 1.0.1" ed. Fox, J.T., Hsu, D.H., Yang, C., 2012. Unobserved Heterogeneity in Matching Games with an Application to Venture Capital. National Bureau of Economic Research.
- Fund, B.R., Pollock, T.G., Baker, T., Wowak, A.J., 2008. Who's the new kid? The process of developing centrality in venture capitalist deal networks. In: *Network Strategy*. Emerald Group Publishing Limited, pp. 563–593.
- Gompers, P., Lerner, J., 1999a. Conflict of interest in the issuance of public securities: Evidence from venture capital. *J. Law Econ.* 42, 1–28.
- Gompers, P.A., 1996. Grandstanding in the venture capital industry. *J. Finan. Econ.* 42, 133–156.
- Gompers, P.A., Lerner, J., 1999b. What drives venture capital fundraising? *Brookings Papers on Economic Activity Microeconomics* 1998, 149–204.
- Gompers, P.A., Mukharlyamov, V., Xuan, Y., 2016. The cost of friendship. *J. Finan. Econ.* 119, 626–644.
- Gu, Q., Lu, X., 2014. Unraveling the mechanisms of reputation and alliance formation: A study of venture capital syndication in China. *Strategic Manage. J.* 35, 739–750.
- Hart, O., Moore, J., 1988. Incomplete contracts and renegotiation. *Econometrica* 56, 755–785.
- Hochberg, Y.V., Lindsey, L.A., Westerfield, M.M., 2015. Resource accumulation through economic ties: Evidence from venture capital. *J. Finan. Econ.* 118, 245–267.
- Hsu, D.H., 2004. What do entrepreneurs pay for venture capital affiliation? *J. Finance* 59, 1805–1844.
- Humphery-Jenner, M., Suchard, J.-A., 2013. Foreign venture capitalists and the internationalization of entrepreneurial companies: Evidence from China. *J. Int. Bus. Studies* 44, 607–621.
- Huyghebaert, N., Priem, R.K., 2015. How do lead financiers select their partners in buyout syndicates? Empirical results from buyout syndicates in Europe. *Eur. Manage. Rev.* 12, 221–246.
- Kräussl, R., Wuebker, R., Schulze, W., 2015. Is Venture Capital Really a Local business? Evidence from US Cross-Border Investment. CFS Working paper.
- Kreps, D.M., Wilson, R., 1982. Reputation and imperfect information. *J. Econ. Theory* 27, 253–279.
- Krishnan, C., Ivanov, V.I., Masulis, R.W., Singh, A.K., 2011. Venture capital reputation, post-IPO performance, and corporate governance. *J. Finan. Quant. Anal.* 46, 1295–1333.
- La Porta, R., Lopez-de-Silanes, F., Shleifer, A., Vishny, R.W., 1998. Law and finance. *J. Pol. Econ.* 106, 1113–1155.
- Lane, H.W., Beamish, P.W., 1990. Cross-cultural cooperative behavior in joint ventures in LDCs. *Manage. Int. Rev.* 30, 87.
- Lee, P.M., Pollock, T.G., Jin, K., 2011. The contingent value of venture capitalist reputation. *Strategic Org.* 9, 33–69.
- Lee, P.M., Wahal, S., 2004. Grandstanding, certification and the underpricing of venture capital backed IPOs. *J. Finan. Econ.* 73, 375–407.
- Lerner, J., 1994. The syndication of venture capital investments. *Finan. Manage.* 16–27.
- Li, J.J., Poppo, L., Zhou, K.Z., 2008. Do managerial ties in China always produce value? Competition, uncertainty, and domestic vs. foreign firms. *Strategic Manage. J.* 29, 383–400.
- LiPuma, J.A., Park, S., 2014. Venture capitalists' risk mitigation of portfolio company internationalization. *Entrepreneurship Theory Pract.* 38, 1183–1205.

- Mäkelä, M.M., Maula, M.V., 2006. Interorganizational commitment in syndicated cross-border venture capital investments. *Entrepreneurship Theory Pract.* 30, 273–298.
- Mäkelä, M.M., Maula, M.V., 2008. Attracting cross-border venture capital: The role of a local investor. *Entrepreneurship Reg. Dev.* 20, 237–257.
- Manigart, S., Wright, M., Lockett, A., Meuleman, M., Landström, H., Bruining, H., Desbrières, P., Hommel, U., 2002. Why do European venture capital companies syndicate? Vlerick Working Papers.
- McPherson, M., Smith-Lovin, L., Cook, J.M., 2001. Birds of a feather: Homophily in social networks. *Ann. Rev. Sociol.* 27, 415–444.
- Meuleman, M., Jääskeläinen, M., Maula, M.V., Wright, M., 2017. Venturing into the unknown with strangers: Substitutes of relational embeddedness in cross-border partner selection in venture capital syndicates. *J. Bus. Venturing* 32, 131–144.
- Meuleman, M., Wright, M., Manigart, S., Lockett, A., 2009. Private equity syndication: Agency costs, reputation and collaboration. *J. Bus. Finance Account.* 36, 616–644.
- Nahata, R., Hazarika, S., Tandon, K., 2014. Success in global venture capital investing: do institutional and cultural differences matter? *J. Financial Quant. Anal.* 49, 1039–1070.
- North, D.C., 1990. *Institutions, Institutional Change and Economic Performance*. Cambridge University Press.
- Petkova, A.P., Wadhwa, A., Yao, X., Jain, S., 2014. Reputation and decision making under ambiguity: A study of US venture capital firms' investments in the emerging clean energy sector. *Acad. Manage. J.* 57, 422–448.
- Plagmann, C., Lutz, E., 2019. Better safe than sorry - the importance of trust for venture capitalists' choice of syndicate partners. *Int. J. Entrepreneurial Venturing* (In press).
- Pruthi, S., Wright, M., Lockett, A., 2003. Do foreign and domestic venture capital firms differ in their monitoring of investees? *Asia Pacific J. Manage.* 20, 175–204.
- Rindova, V.P., Petkova, A.P., Kotha, S., 2007. Standing out: How new firms in emerging markets build reputation. *Strategic Org.* 5, 31–70.
- Rindova, V.P., Williamson, I.O., Petkova, A.P., Sever, J.M., 2005. Being good or being known: An empirical examination of the dimensions, antecedents, and consequences of organizational reputation. *Acad. Manage. J.* 48, 1033–1049.
- Shapiro, C., 1983. Premiums for high quality products as returns to reputations. *Q. J. Econ.* 98, 659–679.
- Sorenson, O., Stuart, T.E., 2001. Syndication networks and the spatial distribution of venture capital investments. *Am. J. Sociol.* 106, 1546–1588.
- Sorenson, O., Stuart, T.E., 2008. Bringing the context back in: Settings and the search for syndicate partners in venture capital investment networks. *Administrative Sci. Q.* 53, 266–294.
- Trapido, D., 2007. Competitive embeddedness and the emergence of interfirm cooperation. *Social Forces* 86, 165–191.
- Tykvová, T., Schertler, A., 2011. Geographical and institutional distances in venture capital deals: How syndication and experience drive internationalization patterns. ZEW-Centre for European Economic Research Discussion Paper.
- Tykvová, T., Schertler, A., 2014. Does syndication with local venture capitalists moderate the effects of geographical and institutional distance? *J. Int. Manage.* 20, 406–420.
- Wilson, R., 1968. The theory of syndicates. *Econometrica* 36, 119–132.
- Wright, M., Lockett, A., 2003. The structure and management of alliances: Syndication in the venture capital industry. *J. Manage. Stud.* 40, 2073–2102.
- Zhelyazkov, P.I., Gulati, R., 2015. After the break-up: The relational and reputational consequences of withdrawals from venture capital syndicates. *Acad. Manage. J.* 59, 277–301.

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